

第六章 卤代烃

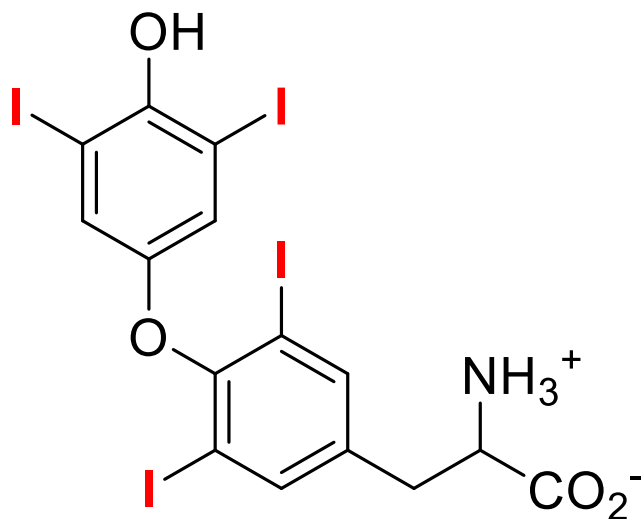
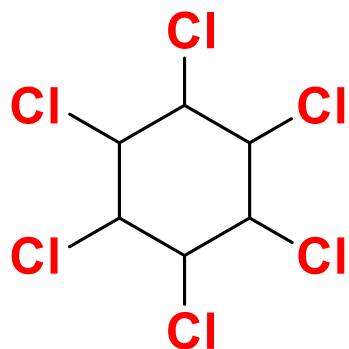
本章内容摘要

6.1 卤代烃的分类、命名和结构

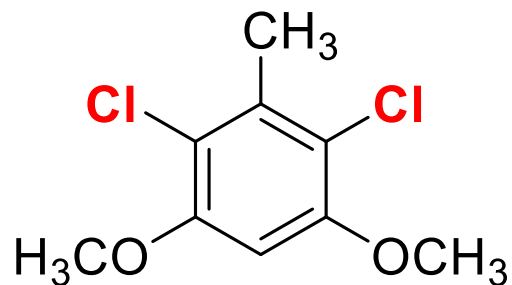
6.2 卤代烃的物理性质

6.3 卤代烃的化学反应

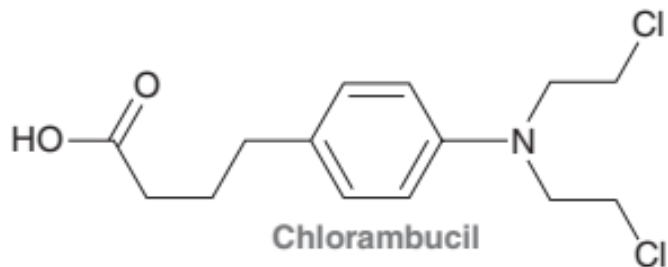
- 亲核取代反应及影响因素
- 消除反应及影响因素
- 亲核取代反应与消除反应的竞争



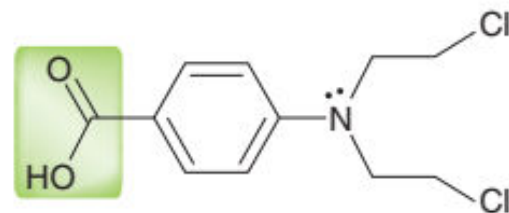
thyroxine (T4) 甲状腺素



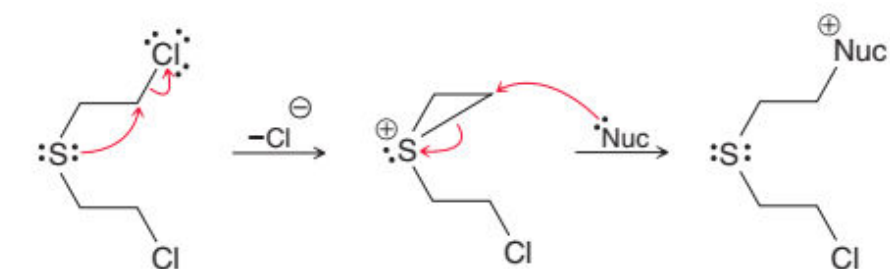
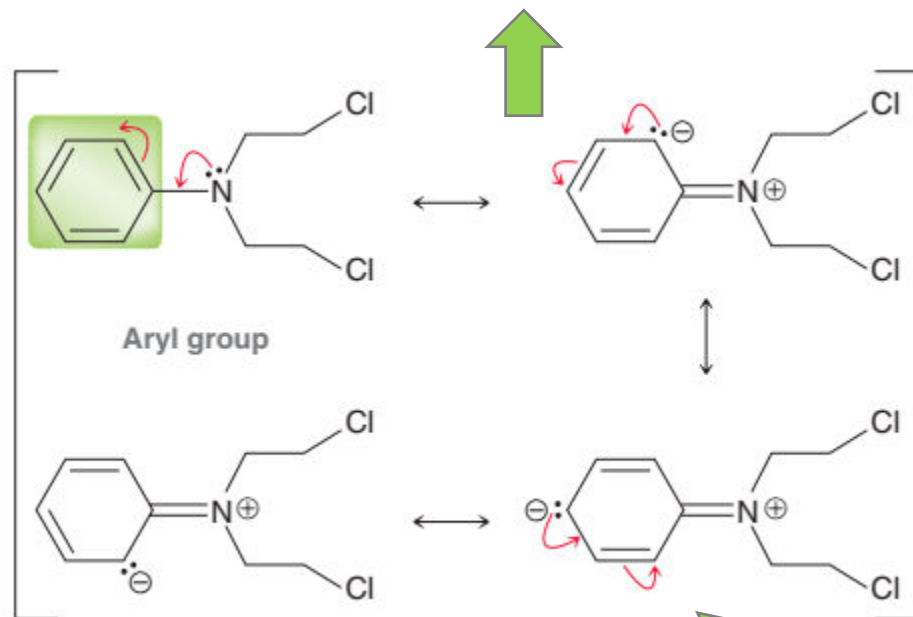
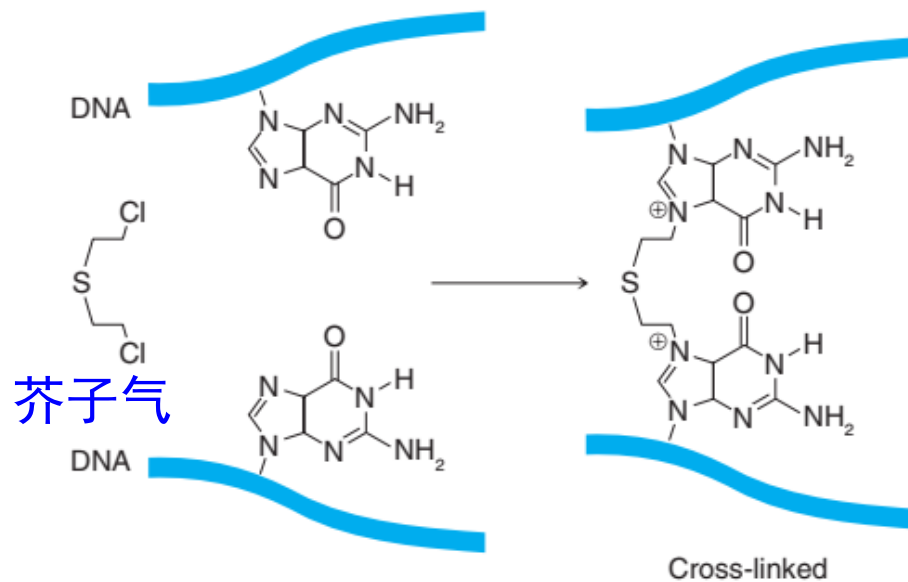
an antifungal compound
isolated from lily plants



药物名：
Lukeran 瘤可宁
CLL白血病药物



This group renders the compound water soluble.



Sulfur mustard

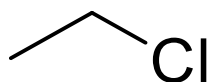
Nitrogen mustard
(mechlorethamine)

6.1 卤代烃的分类、命名和结构

卤代烃：烃分子中一个或多个氢原子被卤原子（F, Cl, Br, I）取代而生成的化合物。

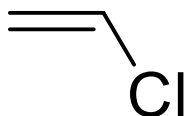
按卤素所连接的烃基类型分为：

- 饱和卤代烃



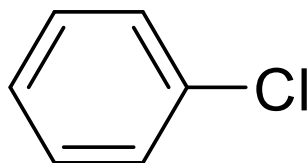
氯乙烷

- 不饱和卤代烃



氯乙烯

- 芳香族卤代烃



氯苯

6.1 卤代烃的分类、命名和结构

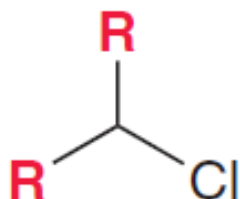
卤代烷烃，按卤素所连接的碳原子类型，分为：

一级、二级、三级卤代烃



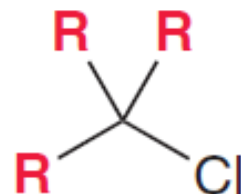
Primary
(1°)

伯卤代物



Secondary
(2°)

仲卤代物



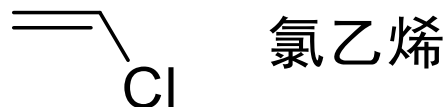
Tertiary
(3°)

叔卤代物

6.1 卤代烃的分类、命名和结构

卤代烯烃的分类

乙烯型卤代烯烃（卤原子直接与双键碳原子相连）



烯丙型卤代烯烃（卤原子直接与双键的 α -碳原子相连）



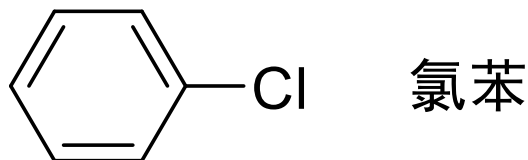
隔离型卤代烯烃（卤原子与距双键相隔两个或两个以上的饱和碳原子相连）



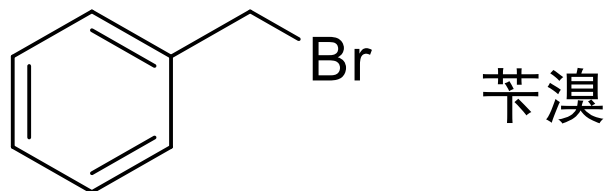
6.1 卤代烃的分类、命名和结构

卤代芳烃的分类

卤苯型卤代芳烃（卤原子直接与苯环相连）



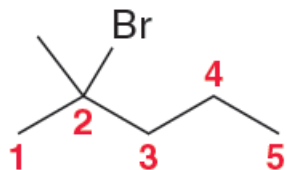
苄型卤代芳烃（卤原子直接与苯环的 α -碳原子相连）



6.1 卤代烃的分类、命名和结构

命名时将卤素视为取代基

- 1、确定主链；
- 2、找到取代基；
- 3、指定取代基编号；
- 4、按原子序数（中文）或字母顺序（英文）排列取代基命名



2-Bromo-2-methylpentane

2-甲基-2-溴戊烷

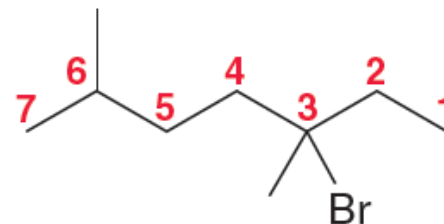
Correct



2, 5, 5 beats 3, 3, 6

2, 5-二甲基-5-溴庚烷

Incorrect

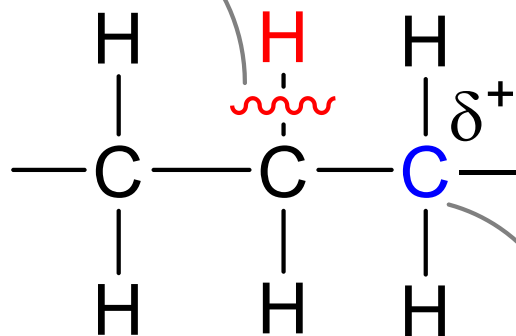


6.2 卤代烃的物理性质

- 室温时氯甲烷、氯乙烷、氯乙烯和溴甲烷为气体，其余为液体， C_{15} 以上为固体。
- 大多数卤代烃具有特殊的气味。
- 除一级氯代烷相对密度小于1外，溴代烷、碘代烷以及多氯代烷的相对密度均大于1。
- 卤代烃难溶于水，易溶于有机溶剂。

6.3 卤代烃的化学反应

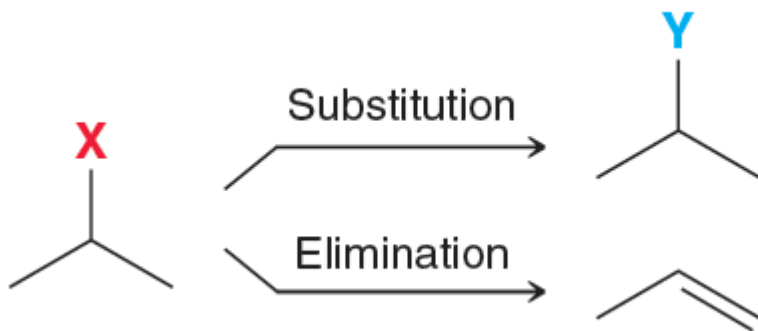
诱导效应，易断裂
“消除”



X吸电子，C-X键键能较小，易断裂，生成的X⁻更稳定。“取代”

带部分正电荷，易被带负电荷或带孤对电子的分子进攻

卤代烷分子由于官能团卤素的存在而具有较活泼的化学性质，反应大多发生在碳卤键上。

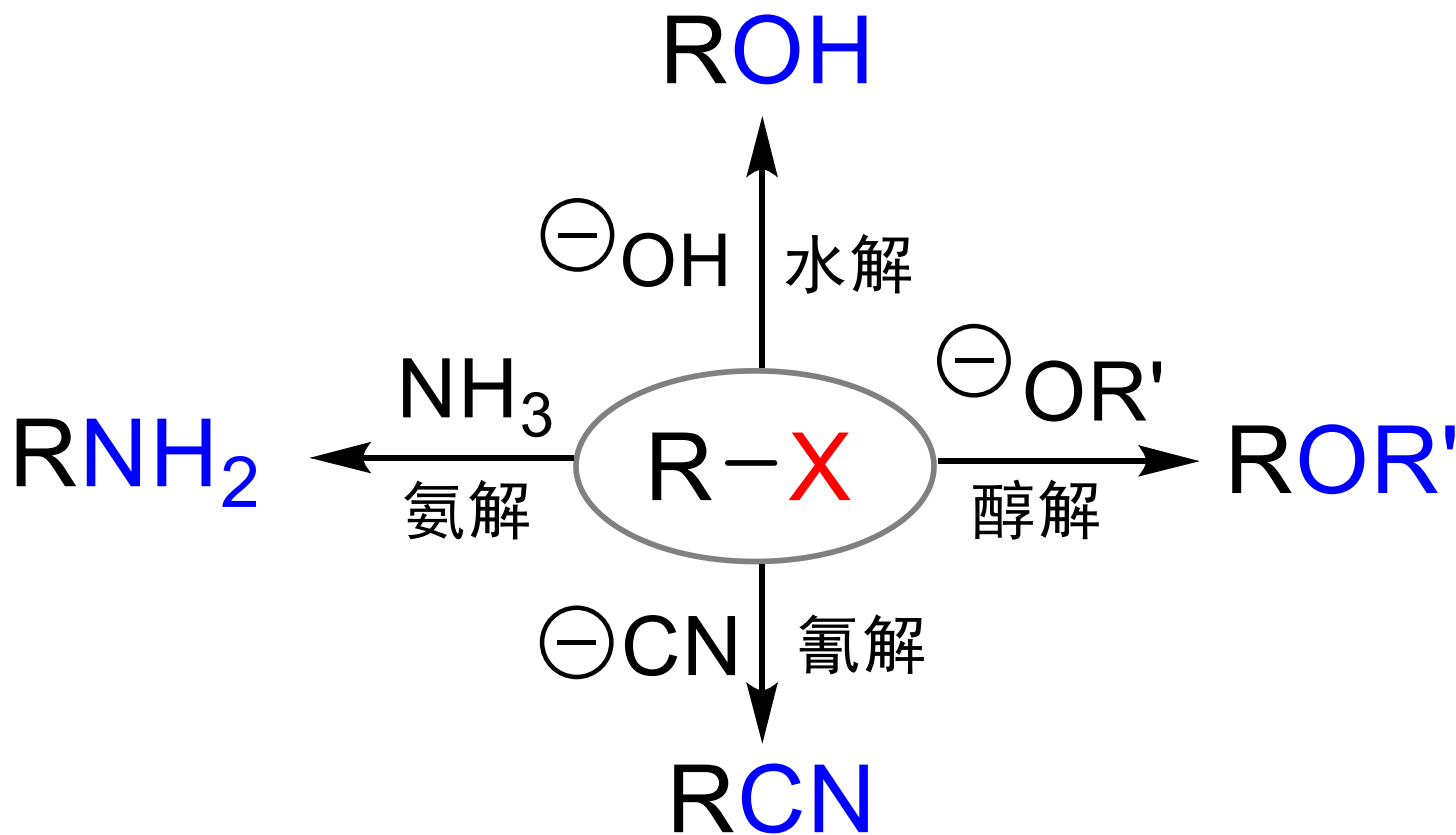


亲核取代反应

消除反应

6.3 卤代烃的化学反应-取代反应

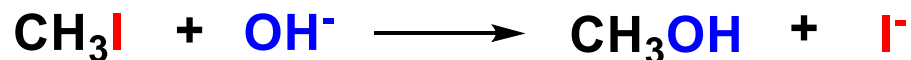
亲核取代反应



6.3 卤代烃的化学反应-取代反应

水解反应

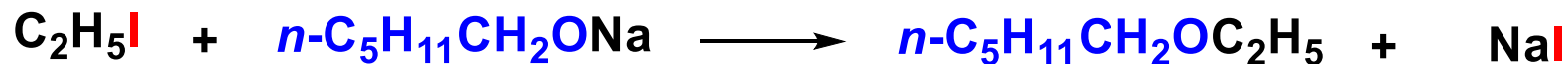
卤代烷与氢氧化钠或氢氧化钾的水溶液一起加热，卤原子被羟基取代生成醇的反应（Hydrolysis）。



6.3 卤代烃的化学反应-取代反应

醇解反应

Williamson合成法：卤代烷与醇钠或醇钾的醇溶液回流，卤原子被烷氧基取代生成醚的反应，该法可以制备混醚（Alcoholysis）。



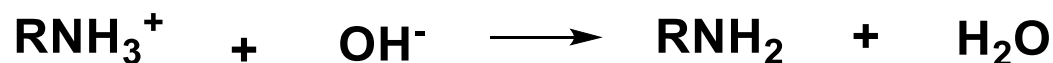
6.3 卤代烃的化学反应-取代反应

氨解反应

卤代烷用浓氨水处理，可以得到伯胺与铵盐的混合物，称为氨解反应。



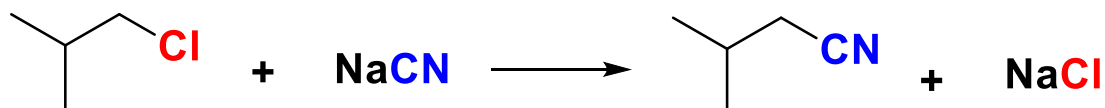
当用碱处理时，可释放出游离胺：



6.3 卤代烃的化学反应-取代反应

氰解反应

用KCN(或NaCN)的乙醇/水溶液（或以DMSO为溶剂）与卤代烷反应，卤原子被氰基取代生成腈的反应。

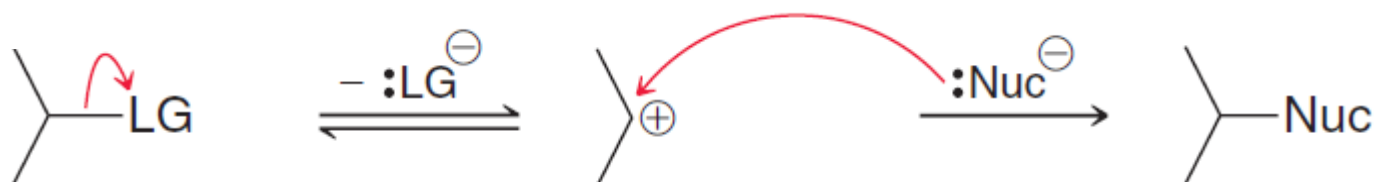


这个反应的产物较原料卤代烷增加了一个碳原子，在有机合成中是增长碳链形成碳碳键的一种方法。引入氰基后可转变为羧基、氨基等。

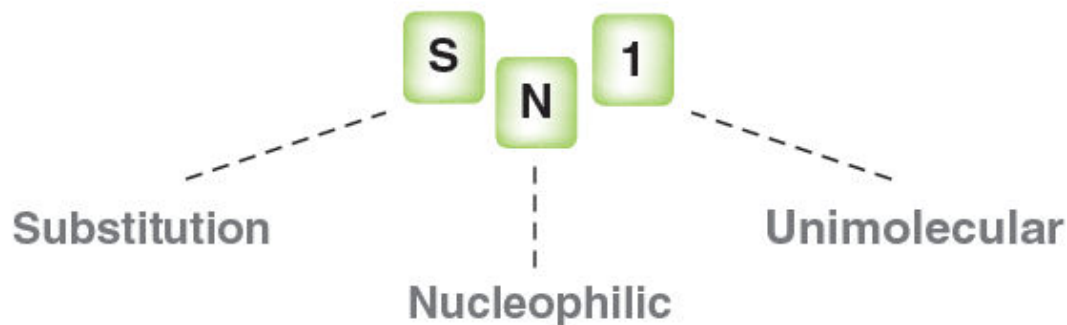
6.3 卤代烃的化学反应-取代反应

卤代烷的亲核取代反应历程

分步机理：先发生基团离去，再进行亲核取代



$$\text{Rate} = k [\text{substrate}]$$

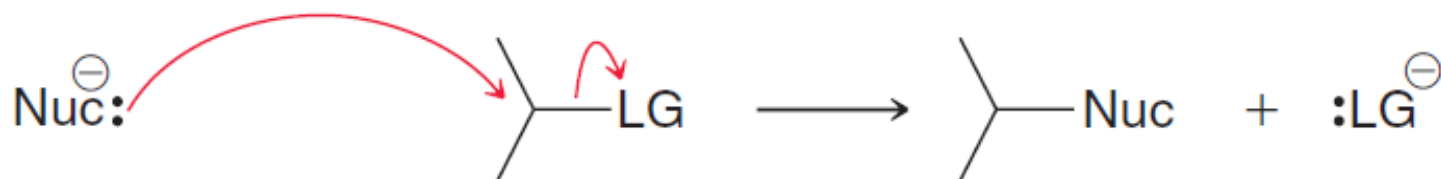


单分子取代反应 S_N1

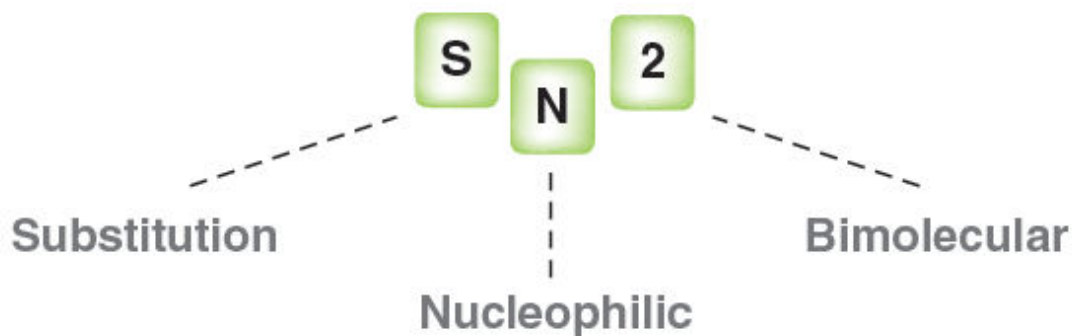
6.3 卤代烃的化学反应-取代反应

卤代烷的亲核取代反应历程

协同机理：亲核进攻与基团离去同时发生



$$\text{Rate} = k [\text{substrate}] [\text{nucleophile}]$$

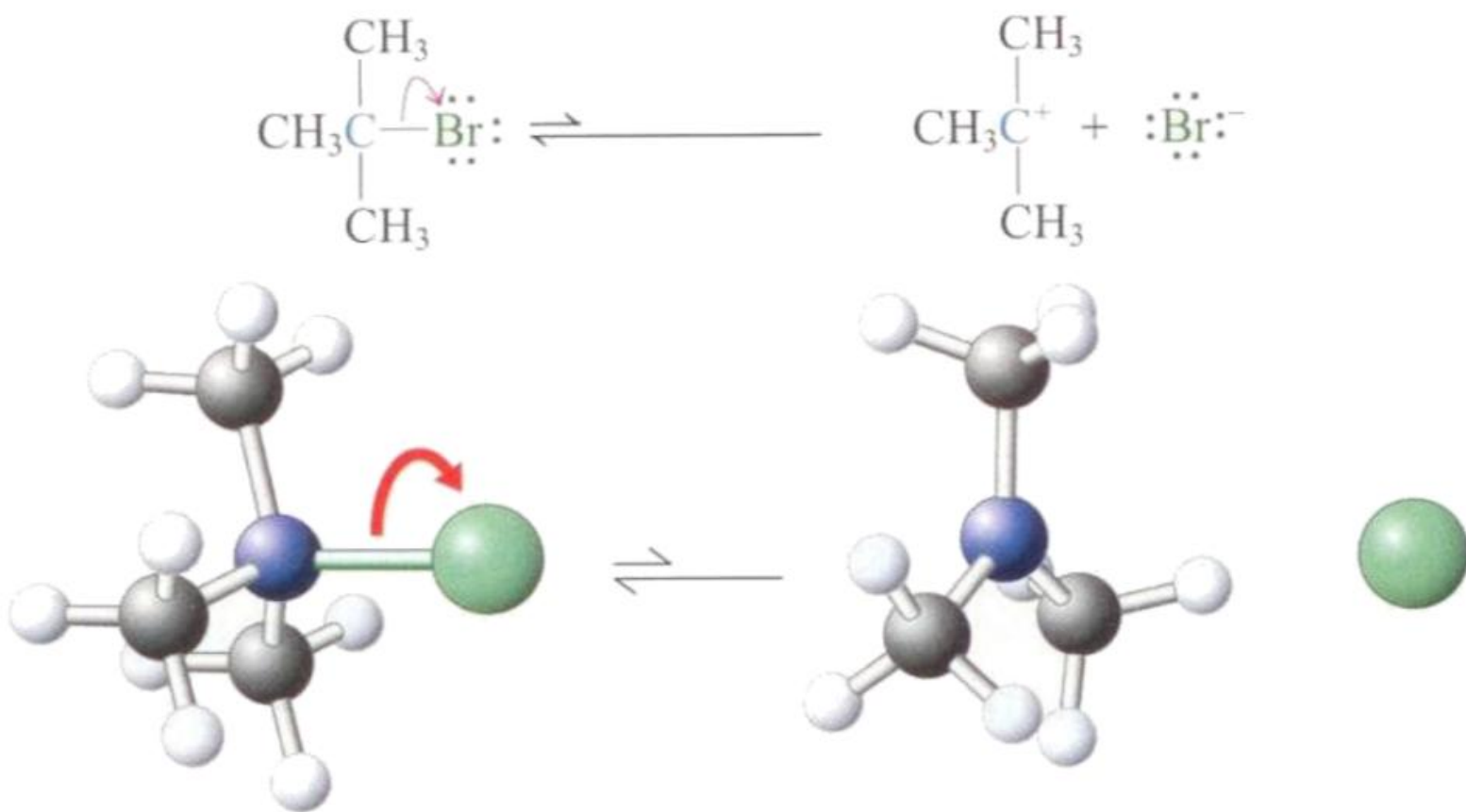


双分子取代反应 $\text{S}_{\text{N}}2$

6.3 卤代烃的化学反应

单分子S_N1取代反应的机理

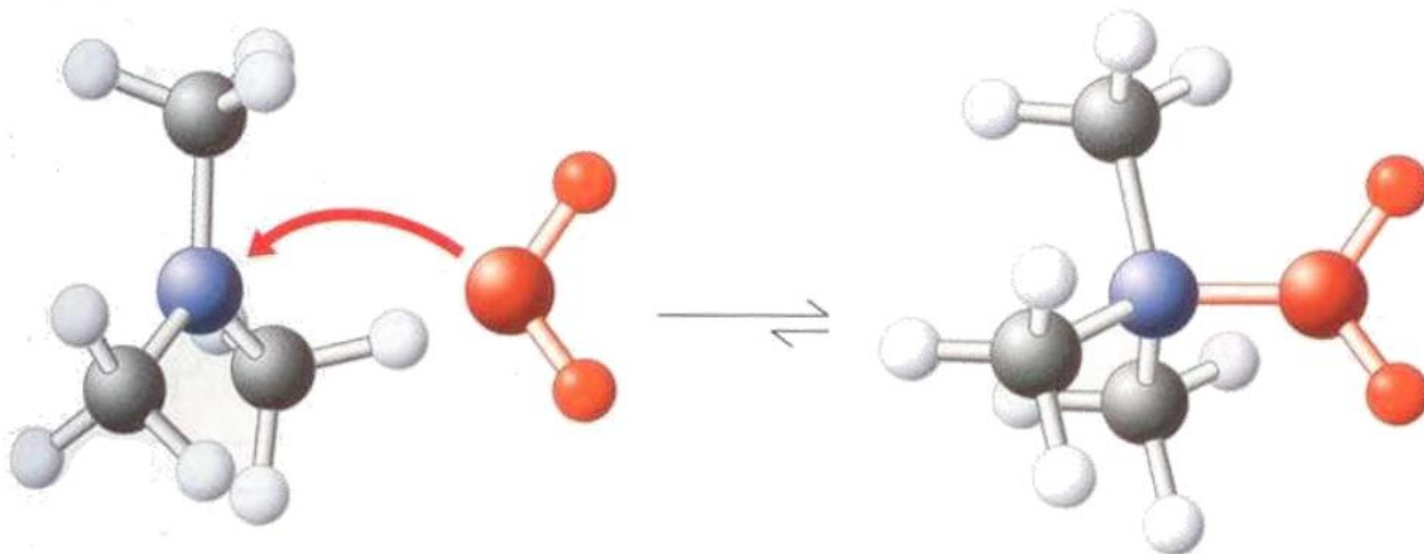
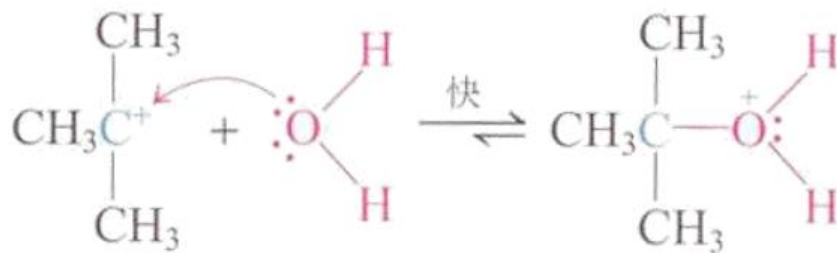
步骤1：卤代烷离解成烷基正离子和卤素负离子（**决速步**）



6.3 卤代烃的化学反应

单分子S_N1取代反应的机理

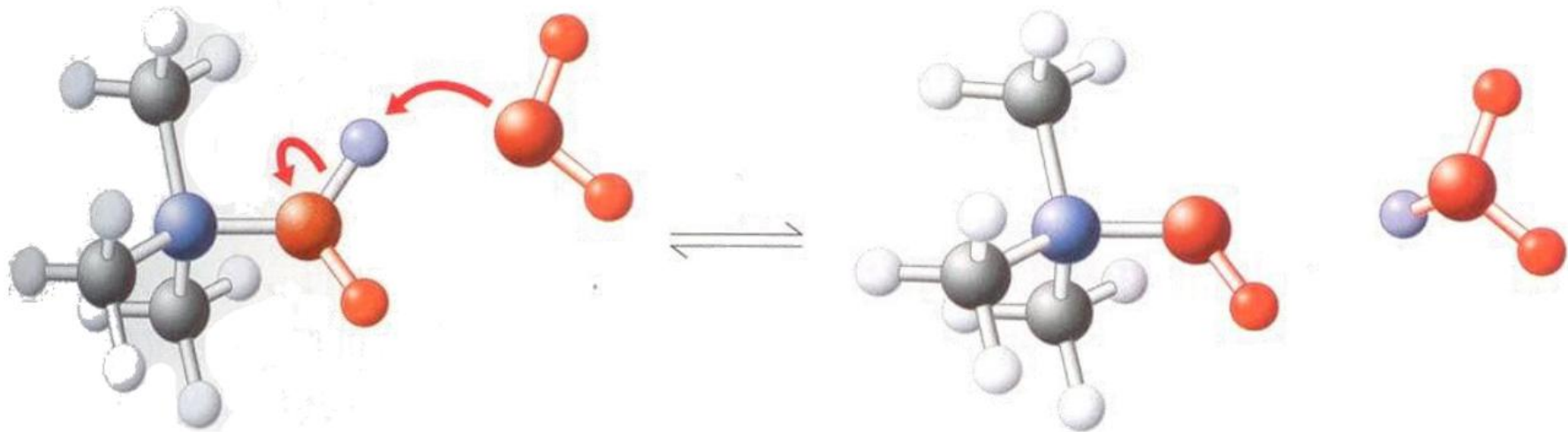
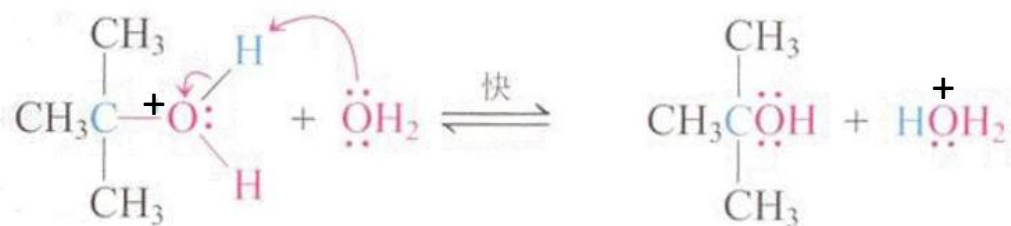
步骤2：水分子亲核进攻烷基正离子



6.3 卤代烃的化学反应-取代反应

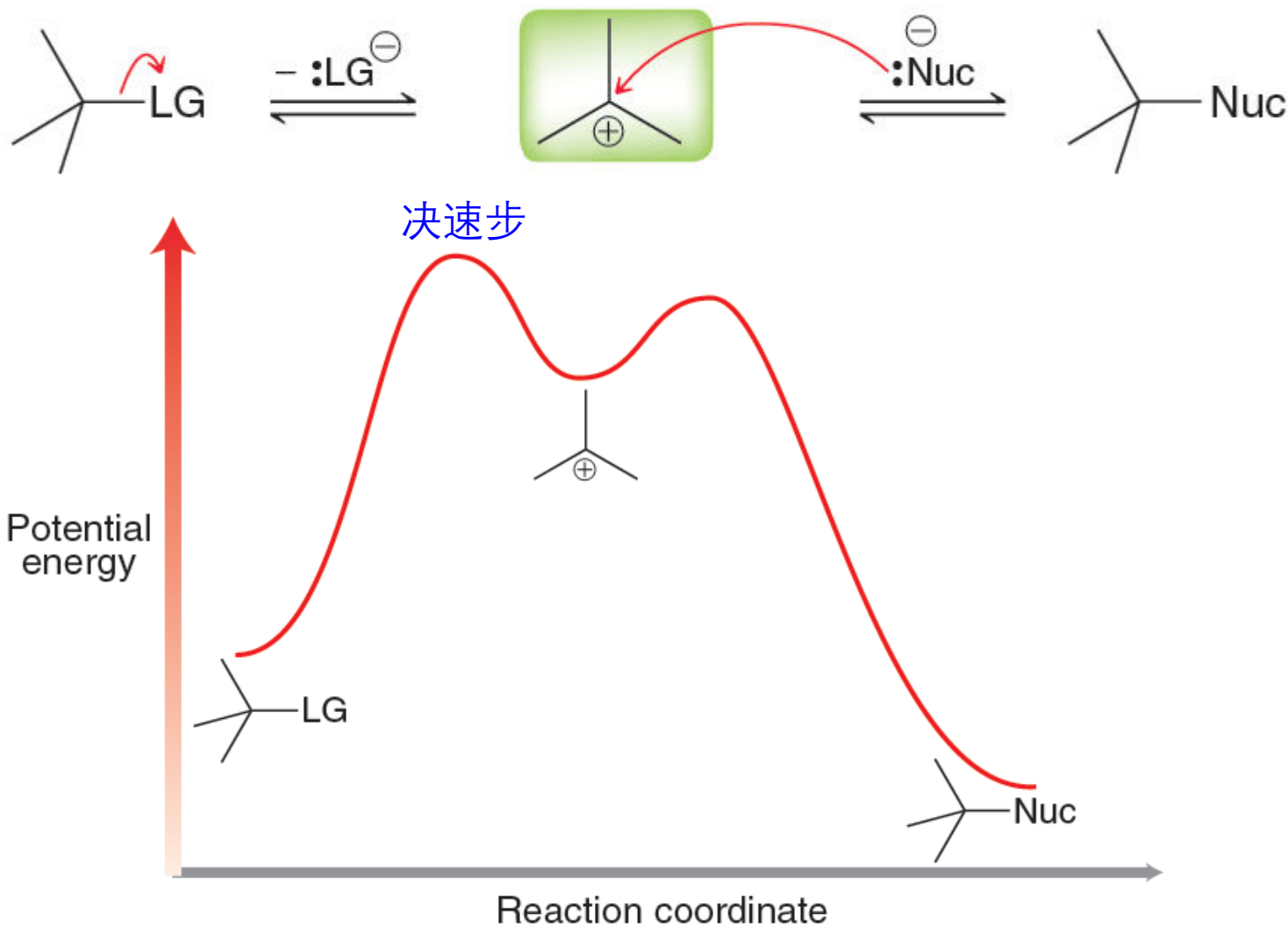
单分子S_N1取代反应的机理

步骤3：质子转移



6.3 卤代烃的化学反应-取代反应

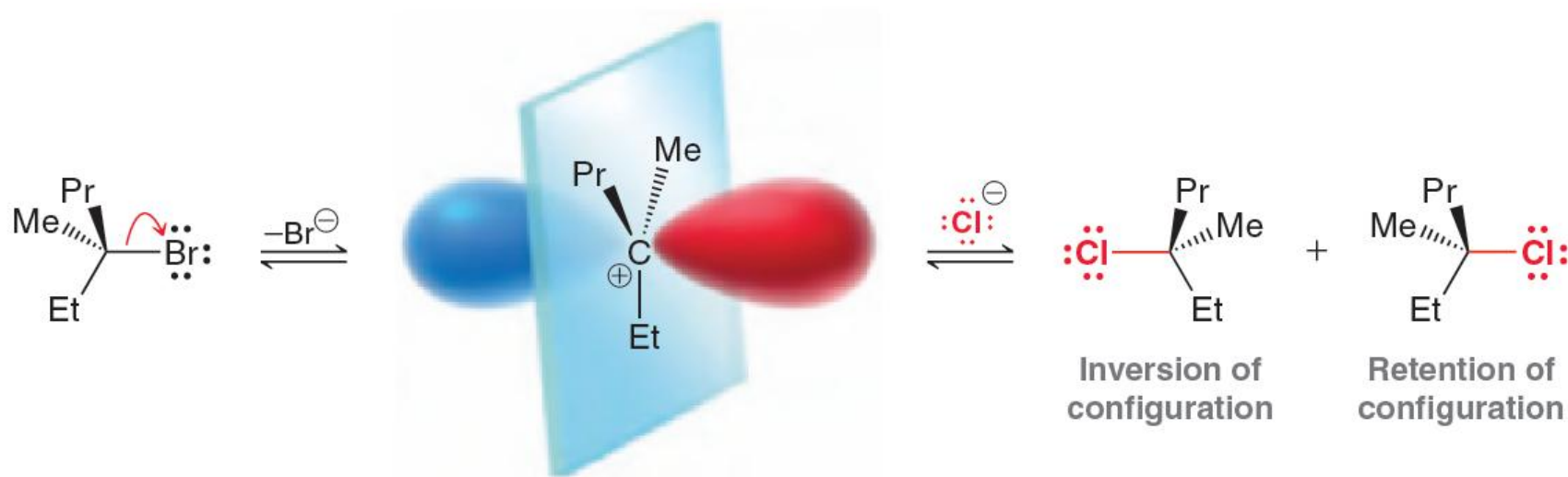
单分子S_N1取代反应的势能面



6.3 卤代烃的化学反应-取代反应

单分子取代反应 S_N1 的特点

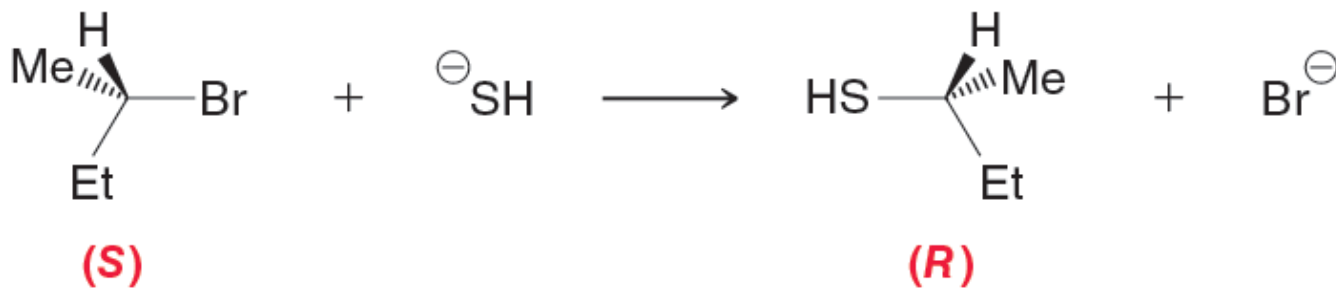
- 反应分步进行，有碳正离子中间体生成，常伴有重排产物；
- 反应速率仅取决于卤代烷的浓度；
- 产物构型一半保持一半翻转。



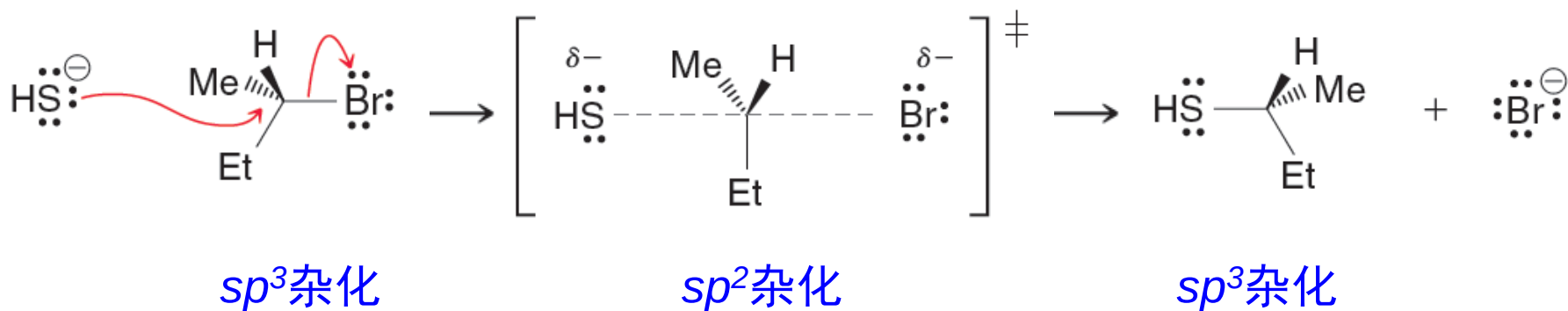
碳正离子为平面构型，两面都可以发生反应，
同时生成立体构型保持及立体构型翻转产物

6.3 卤代烃的化学反应-取代反应

双分子S_N2取代反应的机理



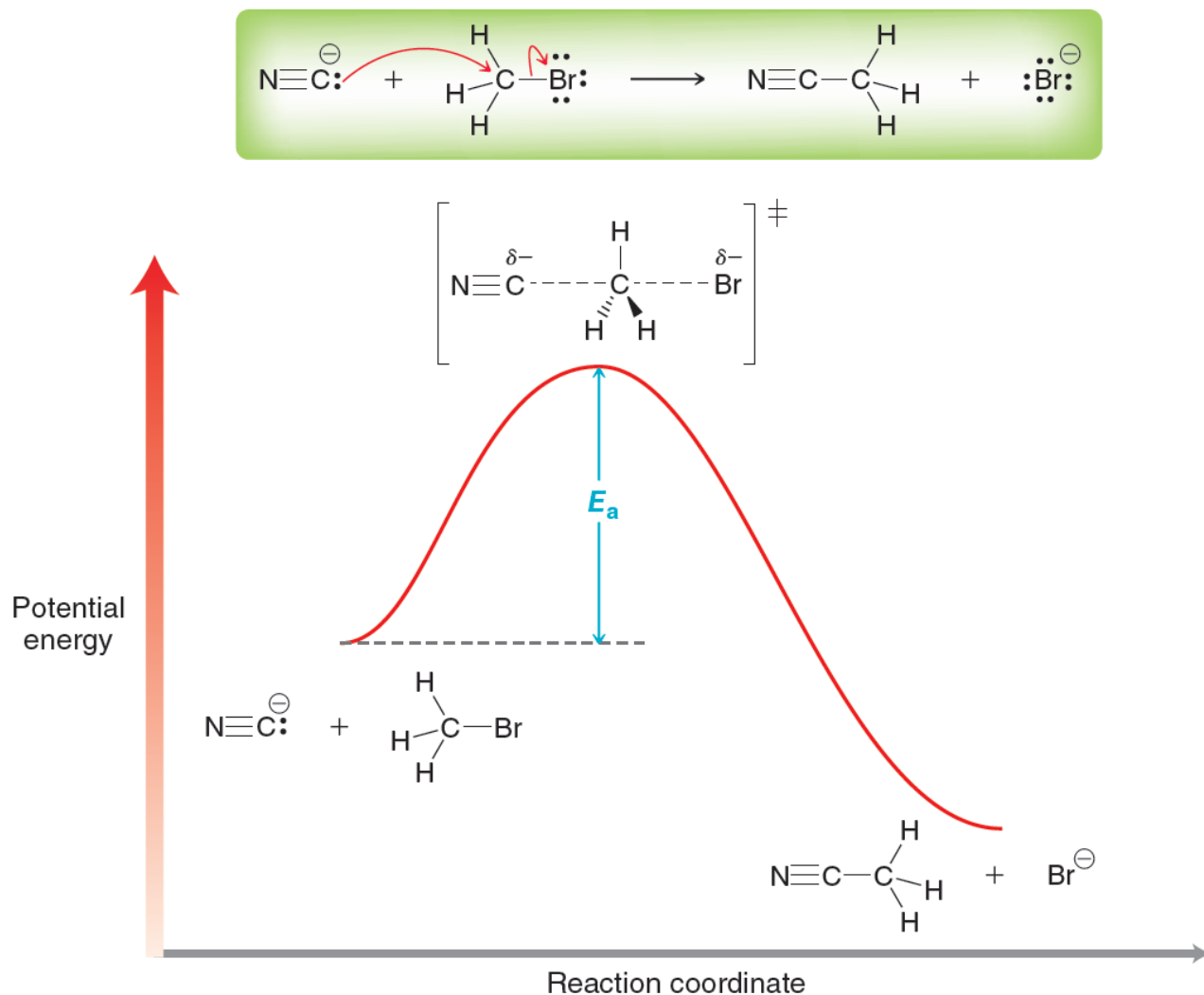
分子构型翻转



亲核试剂从卤原子的背后进攻，生成构型翻转的产物

6.3 卤代烃的化学反应-取代反应

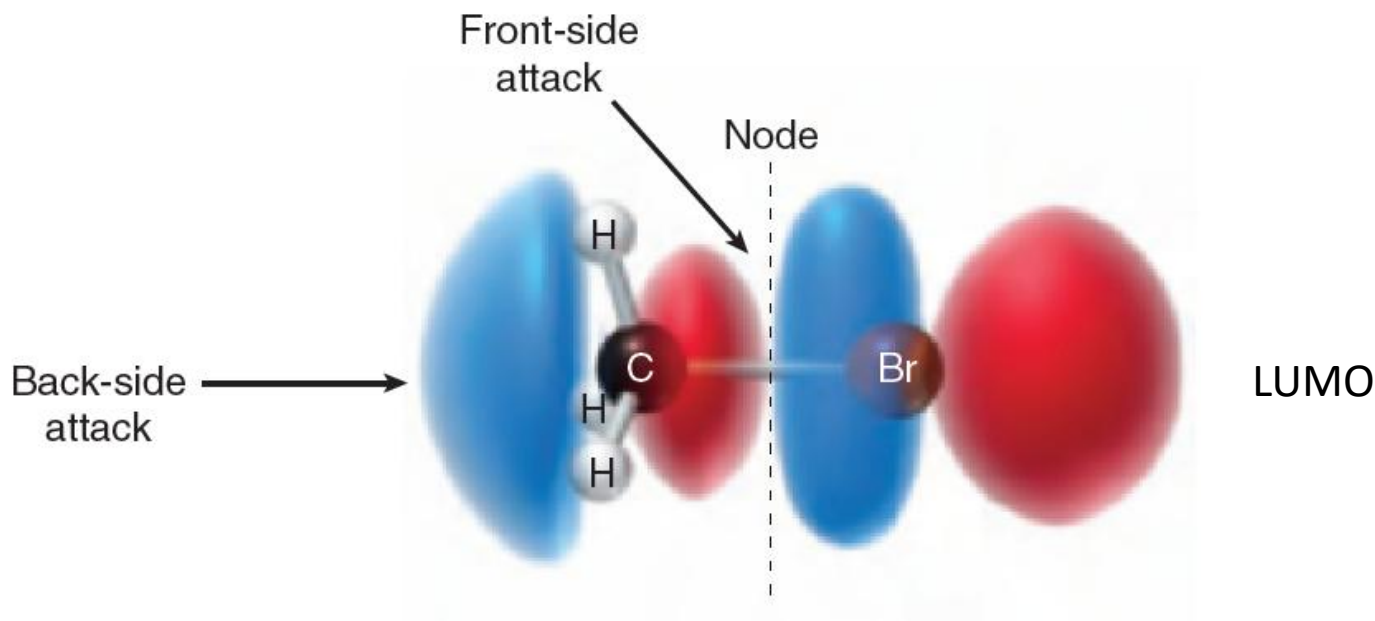
双分子S_N2取代反应的势能面



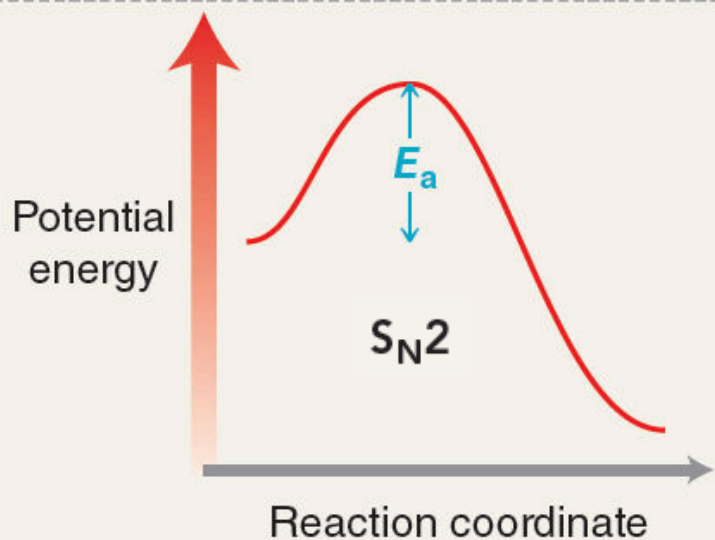
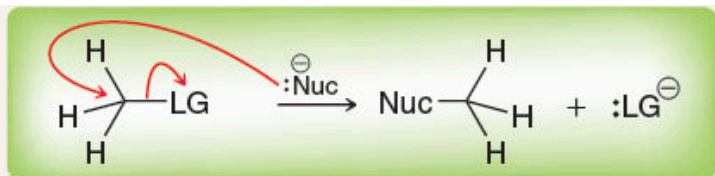
6.3 卤代烃的化学反应-取代反应

双分子取代反应 S_N2 的特点

- 反应连续进行，无中间体生成；
- 反应速率与卤代烷及亲核试剂的浓度成正比；
- 反应前后分子构型发生翻转。



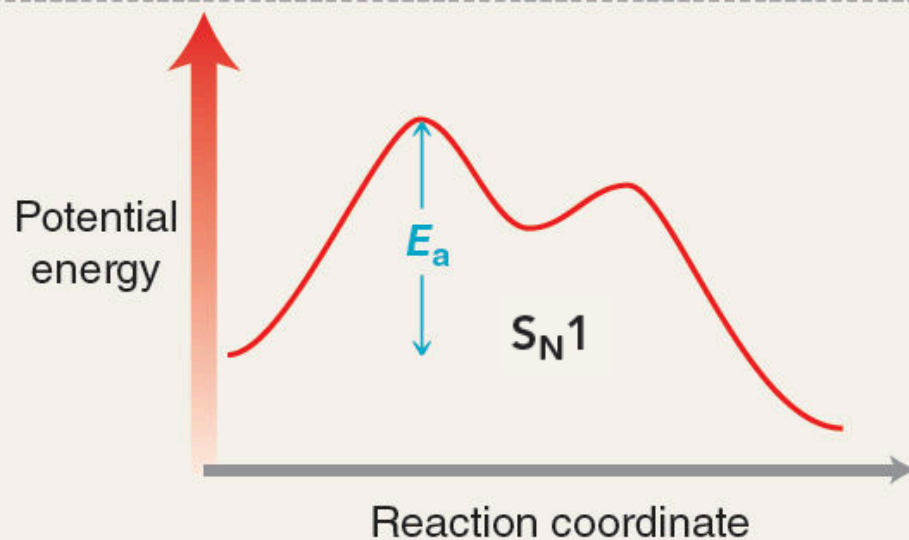
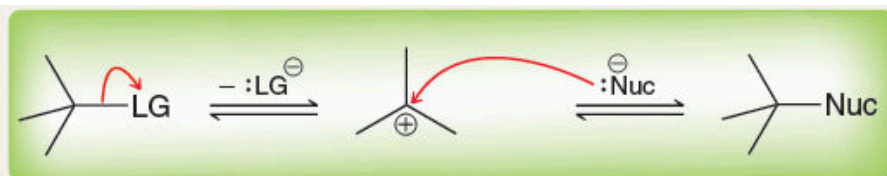
6.3 卤代烃的化学反应-取代反应



反应级数 $\text{Rate} = k [\text{substrate}] [\text{nucleophile}]$

立体构型

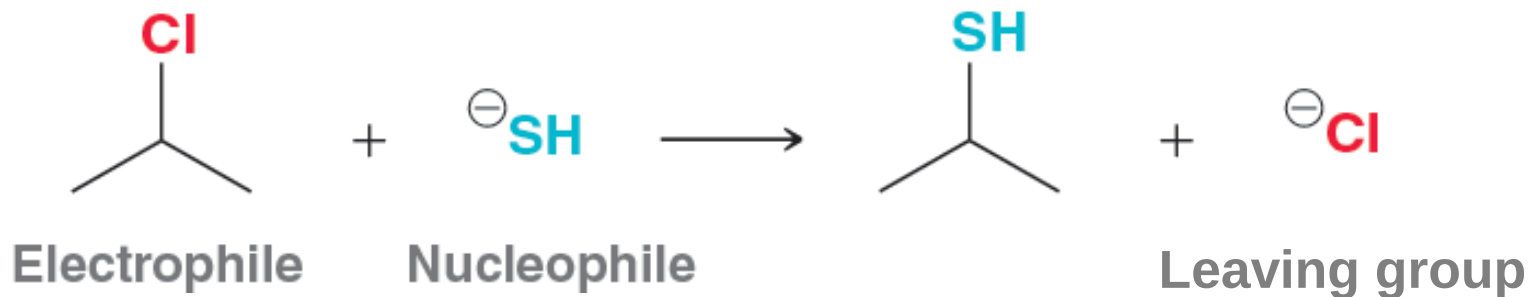
翻转



$\text{Rate} = k [\text{substrate}]$

消旋

6.3 卤代烃的化学反应-取代反应



影响亲核取代反应速率的因素

烷烃结构的影响

离去基团的影响

亲核试剂的影响

溶剂的影响

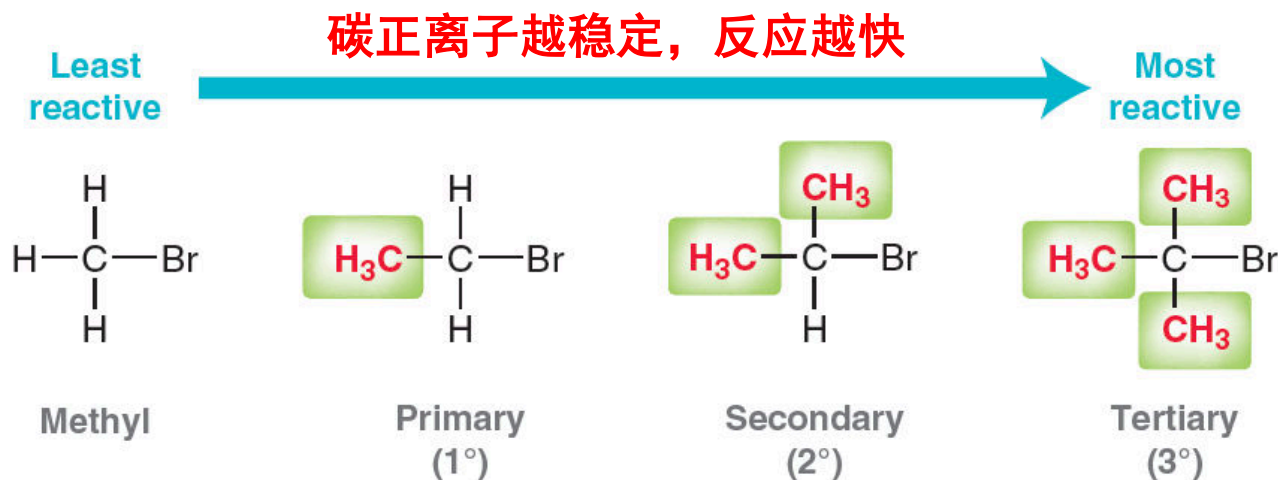
6.3 卤代烃的化学反应-取代反应

烷烃结构的影响

烷基结构对S_N1反应速率的影响

在S_N1反应中，由于决定反应速度的步骤是生成碳正离子一步，而碳正离子的稳定性（σ-p超共轭）顺序为：

三级碳正离子 > 二级碳正离子 > 一级碳正离子 > CH₃⁺



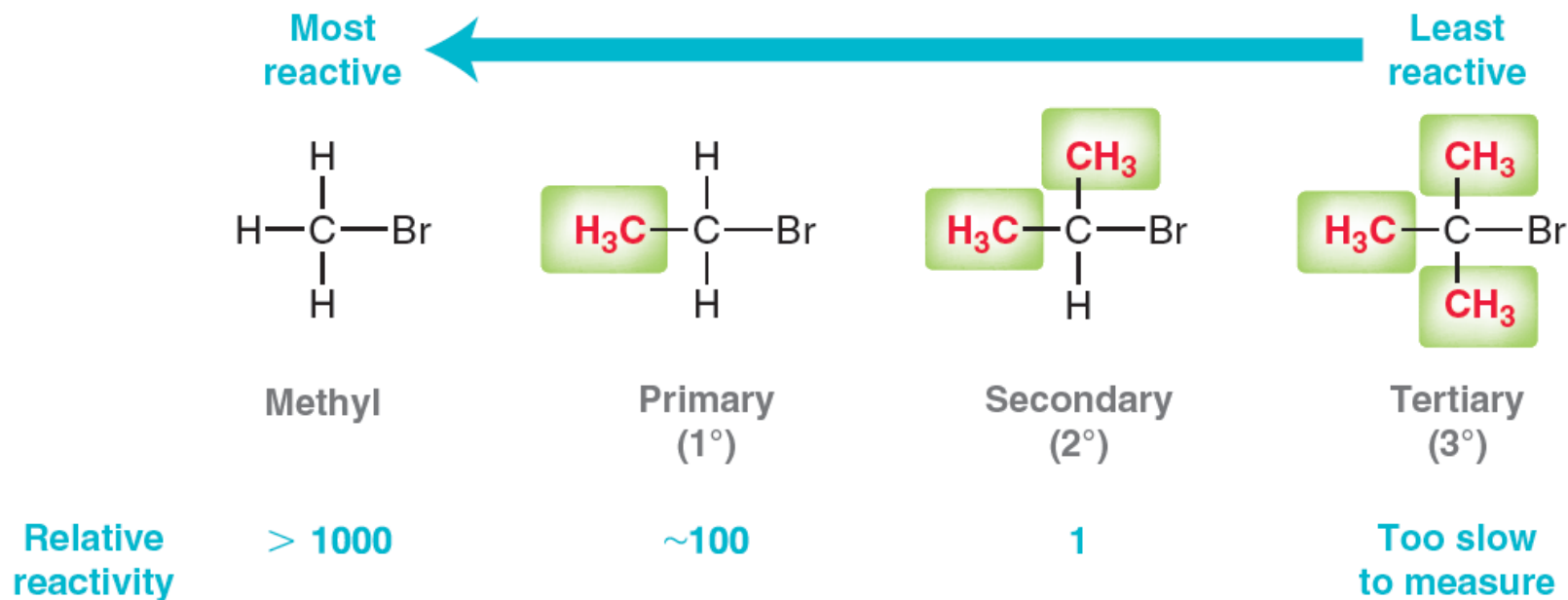
6.3 卤代烃的化学反应-取代反应

烷烃结构的影响

烷基结构对 S_N2 反应速率的影响

在 S_N2 反应中，由于亲核试剂是从卤原子的背后进攻卤代烷的 α -碳原子的，如果 α -碳的取代基增多， α -碳原子周围将变得拥挤，亲核试剂接近 α -碳原子的阻力增加，因此活性也就降低。

位阻越小，反应越快



6.3 卤代烃的化学反应-取代反应

离去基团对亲核取代反应速度的影响

卤代甲烷	键长/Å	键能 /(kJ·mol ⁻¹)
CH ₃ F	1.385	452
CH ₃ Cl	1.794	351
CH ₃ Br	1.929	293
CH ₃ I	2.139	234

C-X键解离能



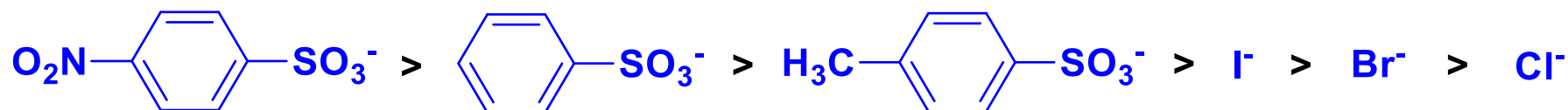
无论S_N1或S_N2反应，都要发生C-X键断裂。卤素的离去倾向越大，反应速率也越快，因此卤代烷的反应活性为：



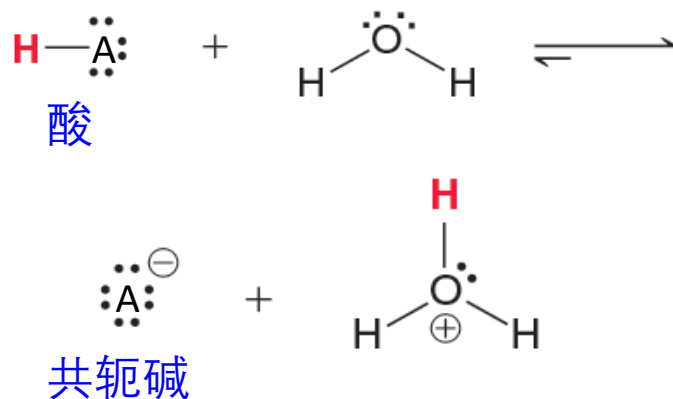
6.3 卤代烃的化学反应-取代反应

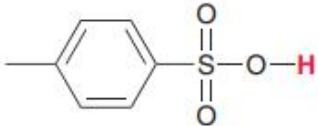
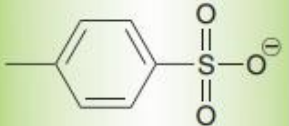
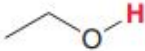
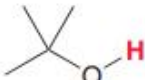
离去基团对亲核取代反应速度的影响

常见的较好的离去基团（能够有效分散负电荷）



强酸的共轭碱是较好的离去基团



Acid	pK _a	Conjugate base
Strongest acid		Most stable base
$\text{I}-\text{H}$	-11	$\text{I}^{\ominus}$
$\text{Br}-\text{H}$	-9	$\text{Br}^{\ominus}$
$\text{Cl}-\text{H}$	-7	$\text{Cl}^{\ominus}$
	-3	
H_3O^+	-2	H_2O
H_2O	15.7	$\text{HO}^{\ominus}$
	16	
	18	
NH_3	38	$\text{NH}_2^{\ominus}$
Weakest acid		Least stable base
		Good leaving groups
		Bad leaving groups

6.3 卤代烃的化学反应-取代反应

亲核试剂对亲核取代反应速度的影响

S_N2 反应速率与亲核试剂强度有关, S_N1 反应速率与亲核试剂无关

- 强亲核试剂有利于 S_N2 反应
- 弱亲核试剂不利于 S_N2 反应 (有利于 S_N1 的竞争)

强亲核试剂			弱亲核试剂
I^-	HS^-	HO^-	F^-
Br^-	H_2S	RO^-	H_2O
Cl^-	RSH	$N\equiv C^-$	ROH

强亲核试剂一般具有较集中的负电荷或较大的极化能力

6.3 卤代烃的化学反应-取代反应

试剂亲核性的判断

- 带有负电荷的亲核试剂比中性化合物亲核性更强



- 含有相同进攻原子或含有同一周期不同进攻原子的亲核试剂，亲核性顺序与碱性性质相同



- 同一主族的亲核试剂，亲核性与碱性顺序相反（离子可极化性）

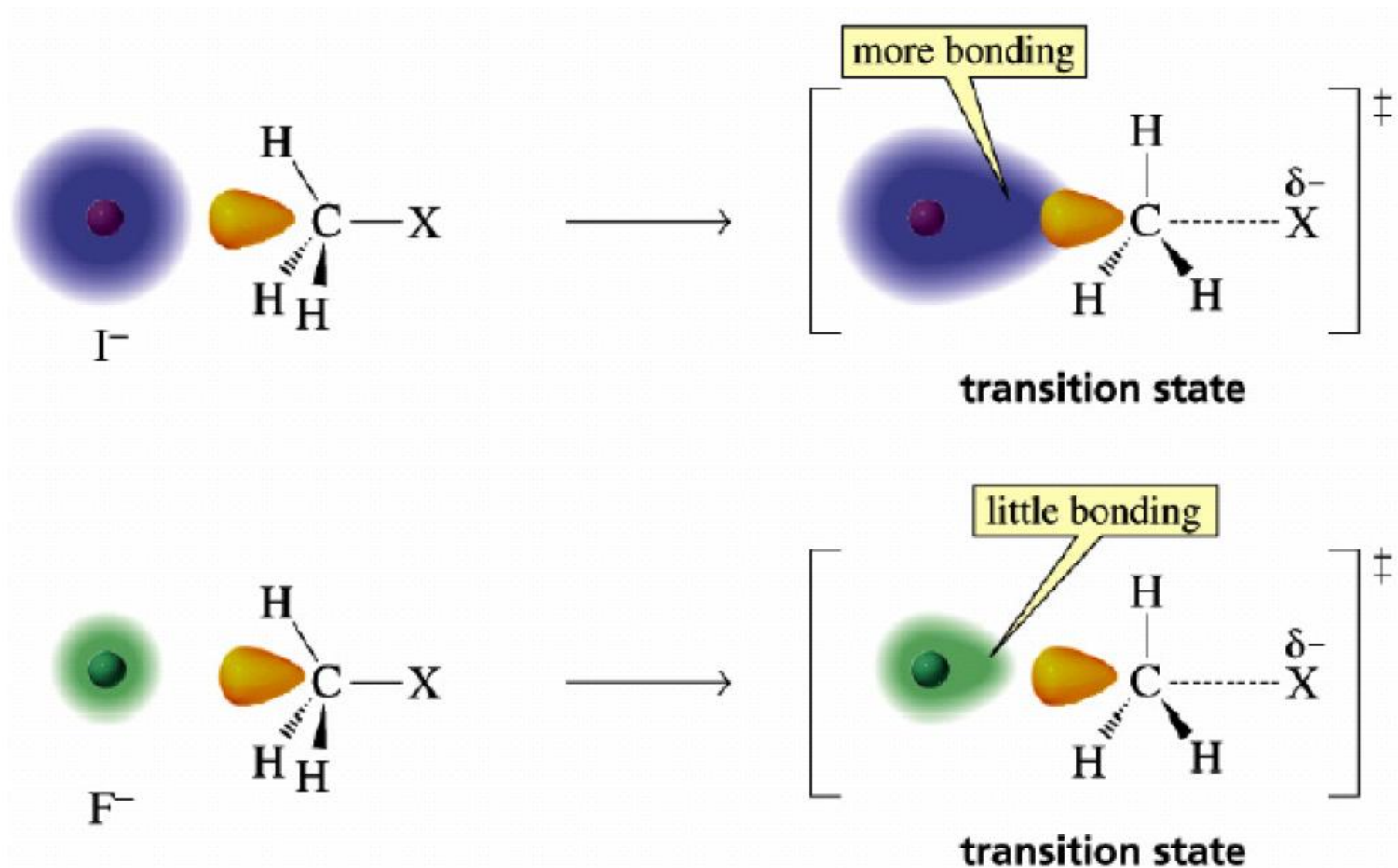


- 体积大，亲核性降低



6.3 卤代烃的化学反应-取代反应

为什么碱性 $\text{F}^- > \text{I}^-$, 而亲核性 $\text{F}^- < \text{I}^-$?



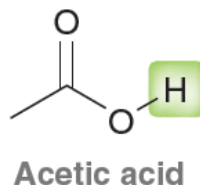
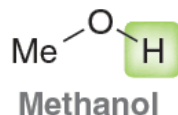
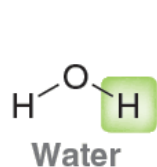
6.3 卤代烃的化学反应-取代反应

溶剂对亲核取代反应速度的影响

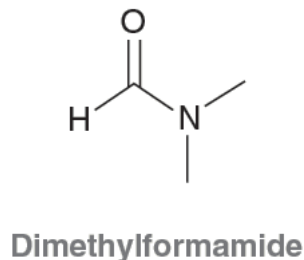
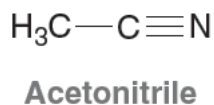
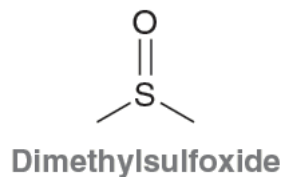
卤代烃的亲核取代反应一般都在极性溶剂中进行。

极性溶剂包括质子性溶剂和非质子性溶剂。

极性质子性溶剂

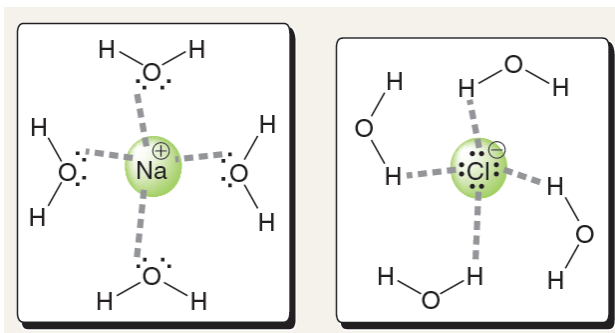


极性非质子性溶剂



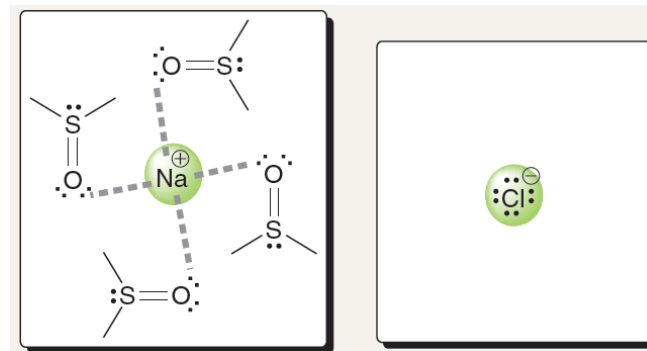
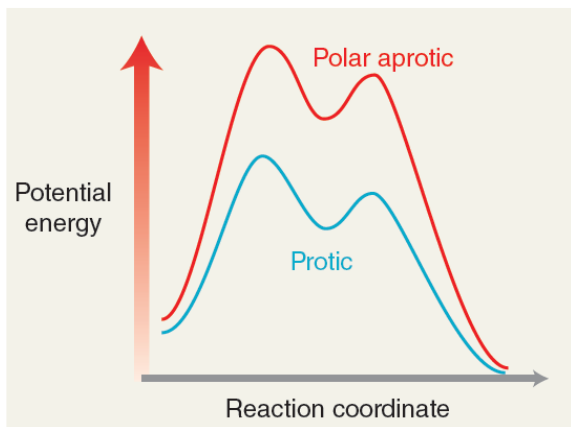
6.3 卤代烃的化学反应-取代反应

溶剂对亲核取代反应速度的影响



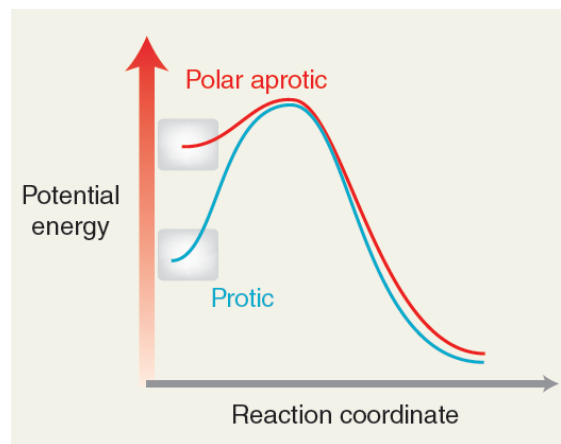
阴阳离子都被溶剂化
对碳正离子起稳定化作用

质子性极性溶剂有利于 S_N1



阳离子被溶剂化
亲核试剂不被溶剂分子包围
亲核试剂能量较高

非质子性极性溶剂有利于 S_N2



LEARN the skill

练习:

Determine whether the following reaction proceeds via an S_N1 or an S_N2 mechanism and then draw the product(s) of the reaction:



SOLUTION

Analyze the four factors one by one:

- (a) *Substrate*. The substrate is secondary. If it were primary, we would predict S_N2 , and if it were tertiary, we would predict S_N1 . But with a secondary substrate, it could be either, so we move on to the next factor.
- (b) *Nucleophile*. NaSH indicates that the nucleophile is HS^- (remember that Na^+ is just the counter ion). HS^- is a strong nucleophile, which favors S_N2 .
- (c) *Leaving Group*. Br^- is a good leaving group. This factor alone does not indicate a preference for either S_N1 or S_N2 .
- (d) *Solvent*. DMSO is a polar aprotic solvent, which favors S_N2 .

Weighing all four factors, there is a preference for S_N2 because both the nucleophile and the solvent favor S_N2 . Therefore, we expect inversion of configuration:



6.3 卤代烃的化学反应-消除反应

卤代烷的消除反应

卤代烷消除反应及查依采夫规则

单分子消除反应历程 (E1)

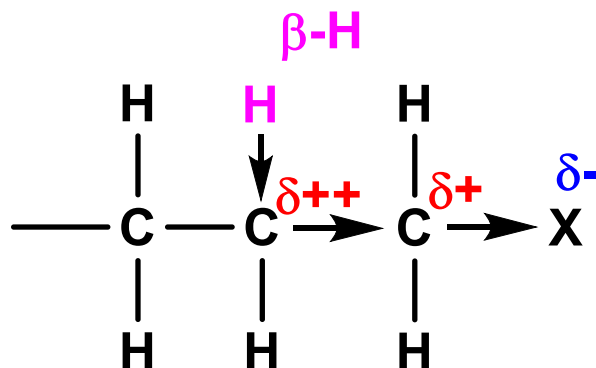
双分子消除反应历程 (E2)

亲核取代反应和消除反应的竞争

6.3 卤代烃的化学反应-消除反应

卤代烷的消除反应

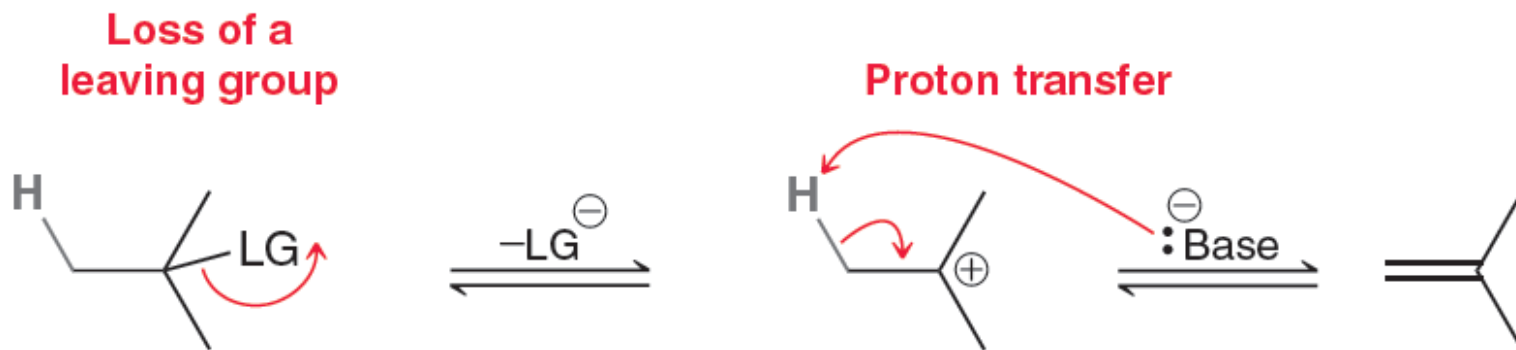
卤代烷与强碱的醇溶液共热，脱去卤原子和相邻碳上的氢原子（ β -H）生成烯烃。这种分子中失去一个简单分子形成不饱和键的反应称为消除反应（Elimination reaction）



6.3 卤代烃的化学反应-消除反应

卤代烷的消除反应类型

分步机理：基团离去先于质子转移（E1）



$$\text{Rate} = k [\text{substrate}]$$

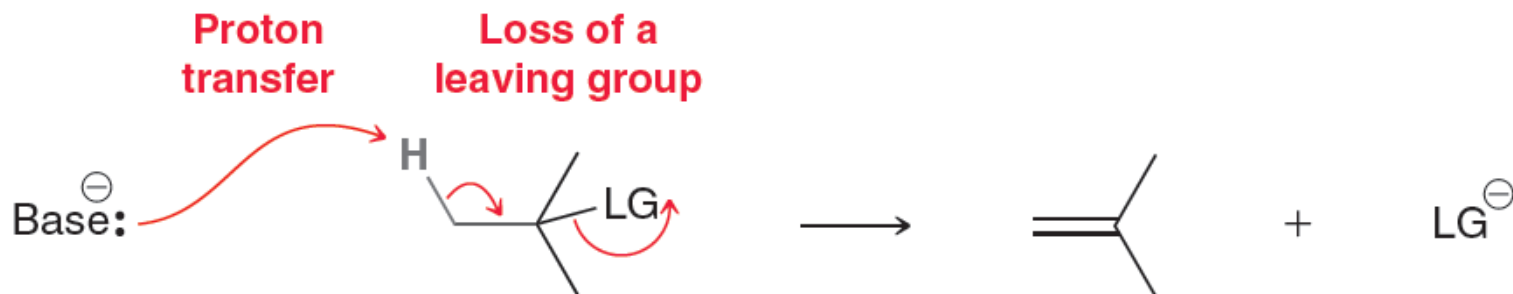


E1与S_N1反应历程相似，多步反应，E1反应中第二步不是正碳离子与亲核试剂结合生成取代产物，而是正碳离子脱去β-H生成消除产物。

6.3 卤代烃的化学反应-消除反应

卤代烷的消除反应类型

协同机理： 质子转移与基团离去同时发生 (E2)

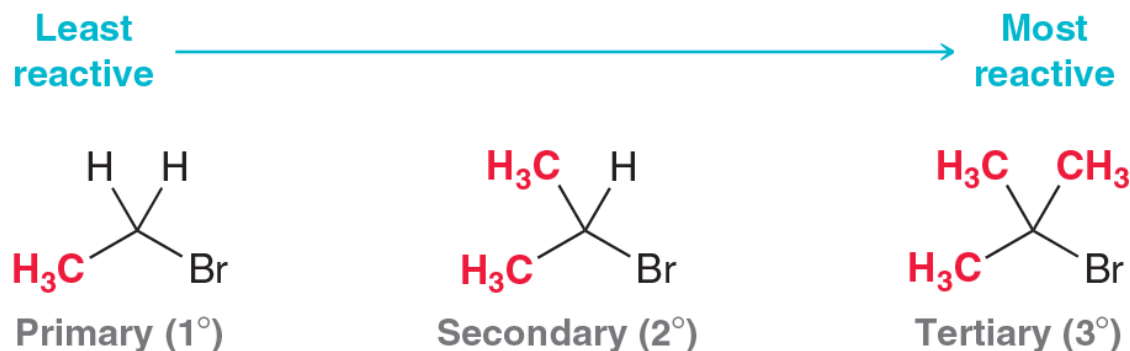


$$\text{Rate} = k [\text{substrate}] [\text{base}]$$

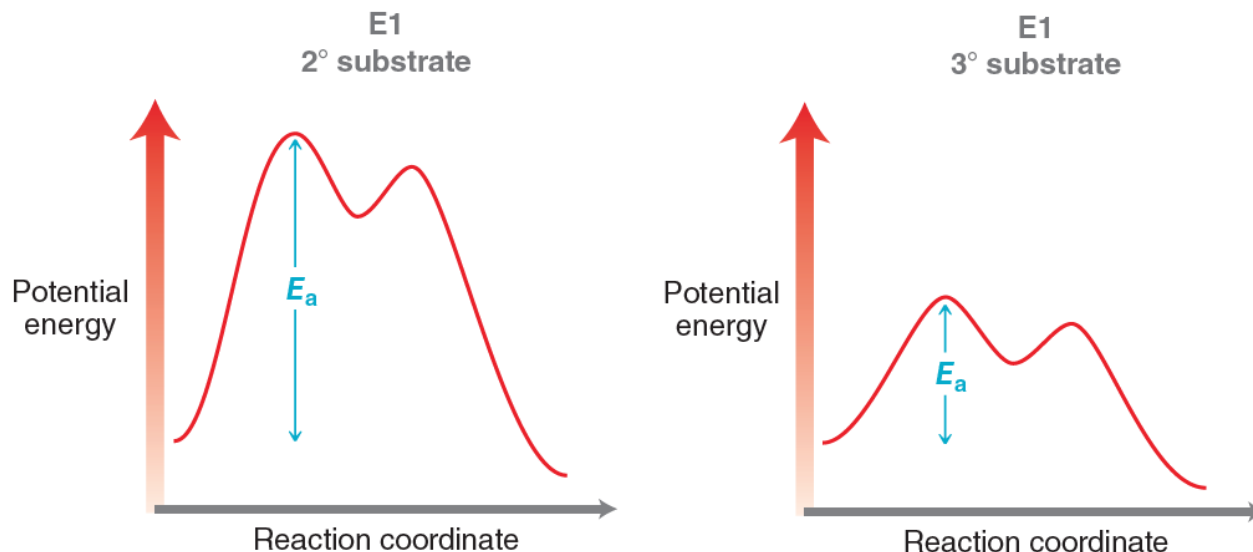


6.3 卤代烃的化学反应-消除反应

单分子消除历程 (E1)



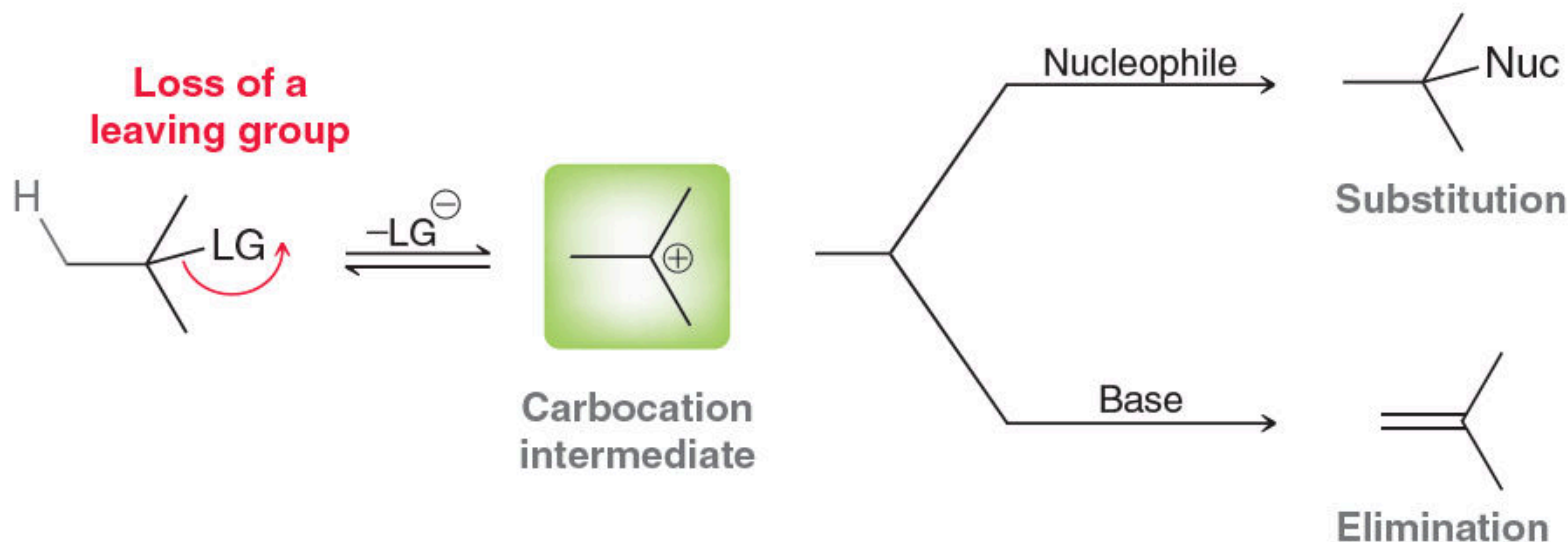
E1反应生成碳正离子中间体，反应的难易程度与碳正离子稳定性有关。碳正离子越稳定，反应越容易发生。



6.3 卤代烃的化学反应-消除反应

E1与S_N1的竞争

E1与S_N1为竞争反应，通常得到二者的混合物

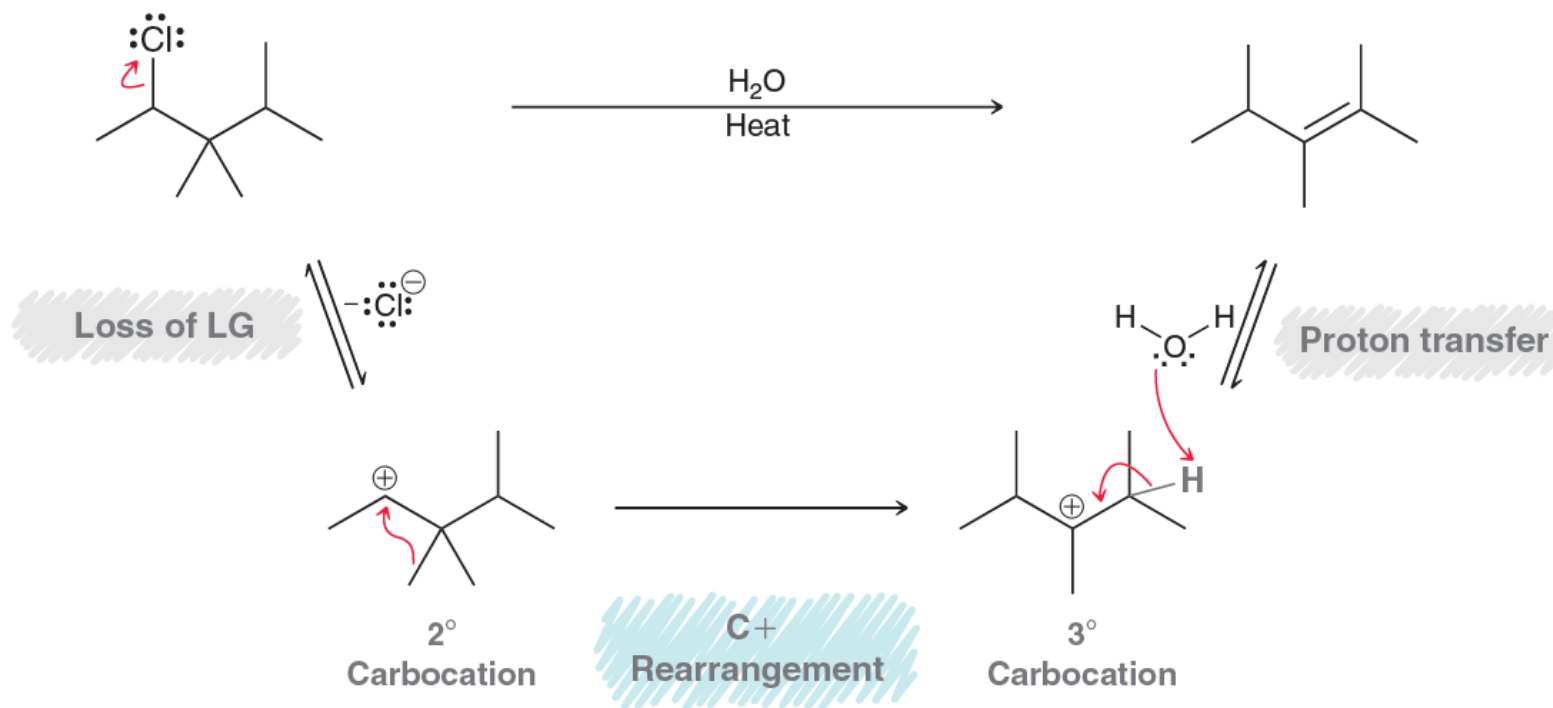


二者经过相同的中间体

6.3 卤代烃的化学反应-消除反应

单分子消除历程 (E1)

如何画反应机理？

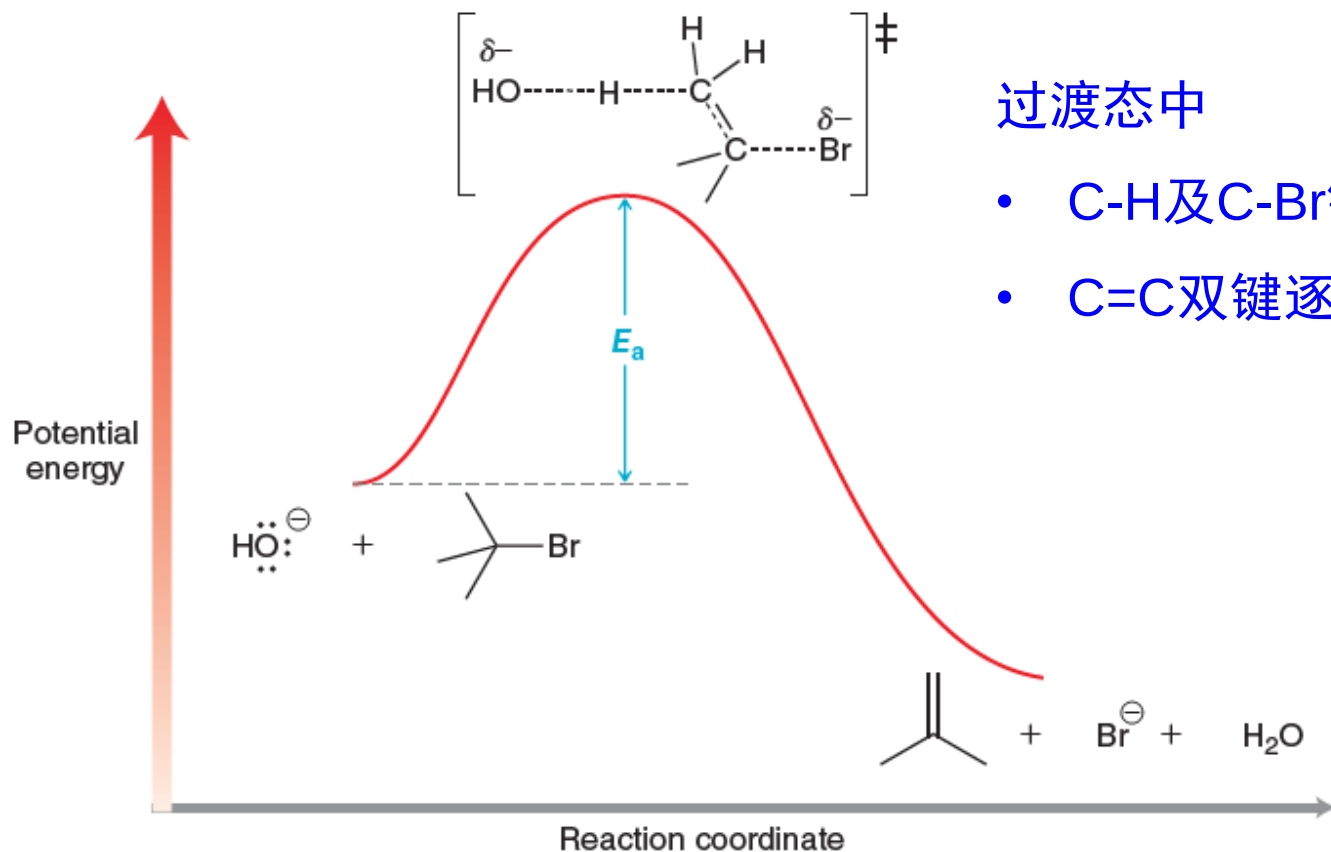
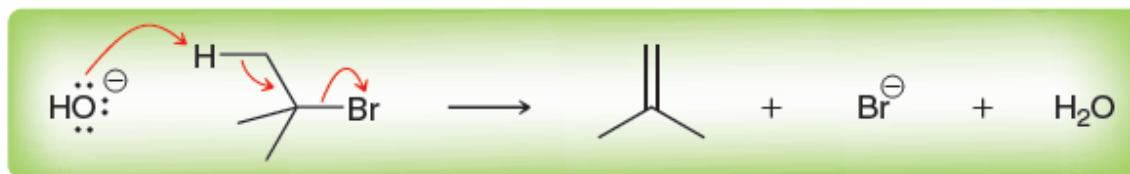


2 core steps, 1 additional step

生成的碳正离子还可以发生重排而转变为更稳定的碳正离子

6.3 卤代烃的化学反应-消除反应

双分子消除历程 (E2)

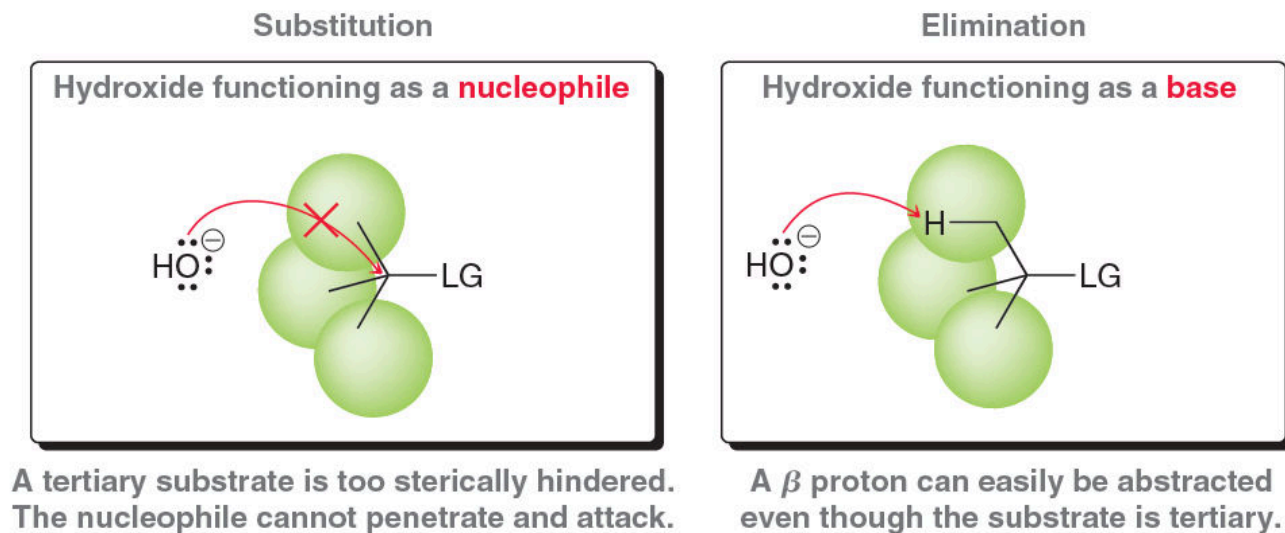
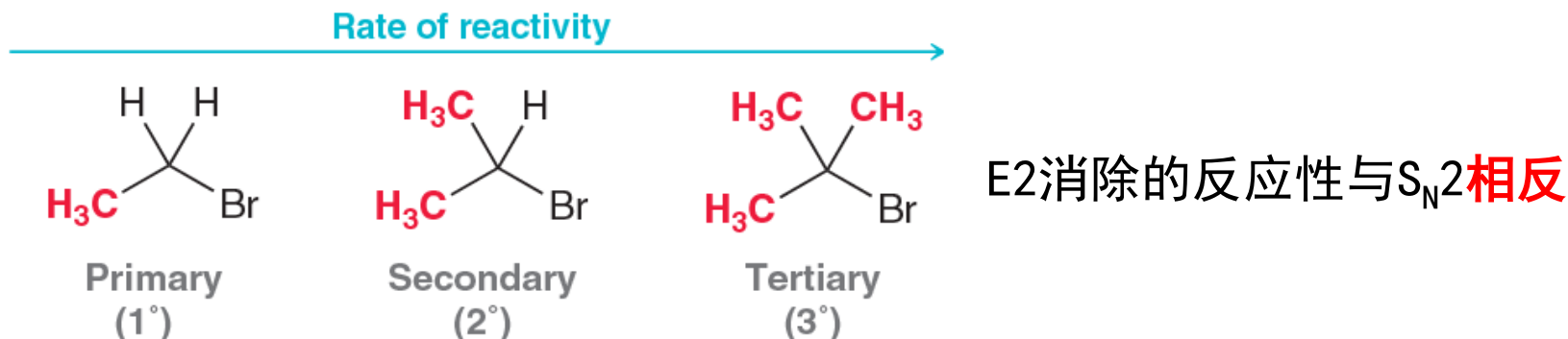


过渡态中

- C-H及C-Br键逐渐断裂
- C=C双键逐渐形成

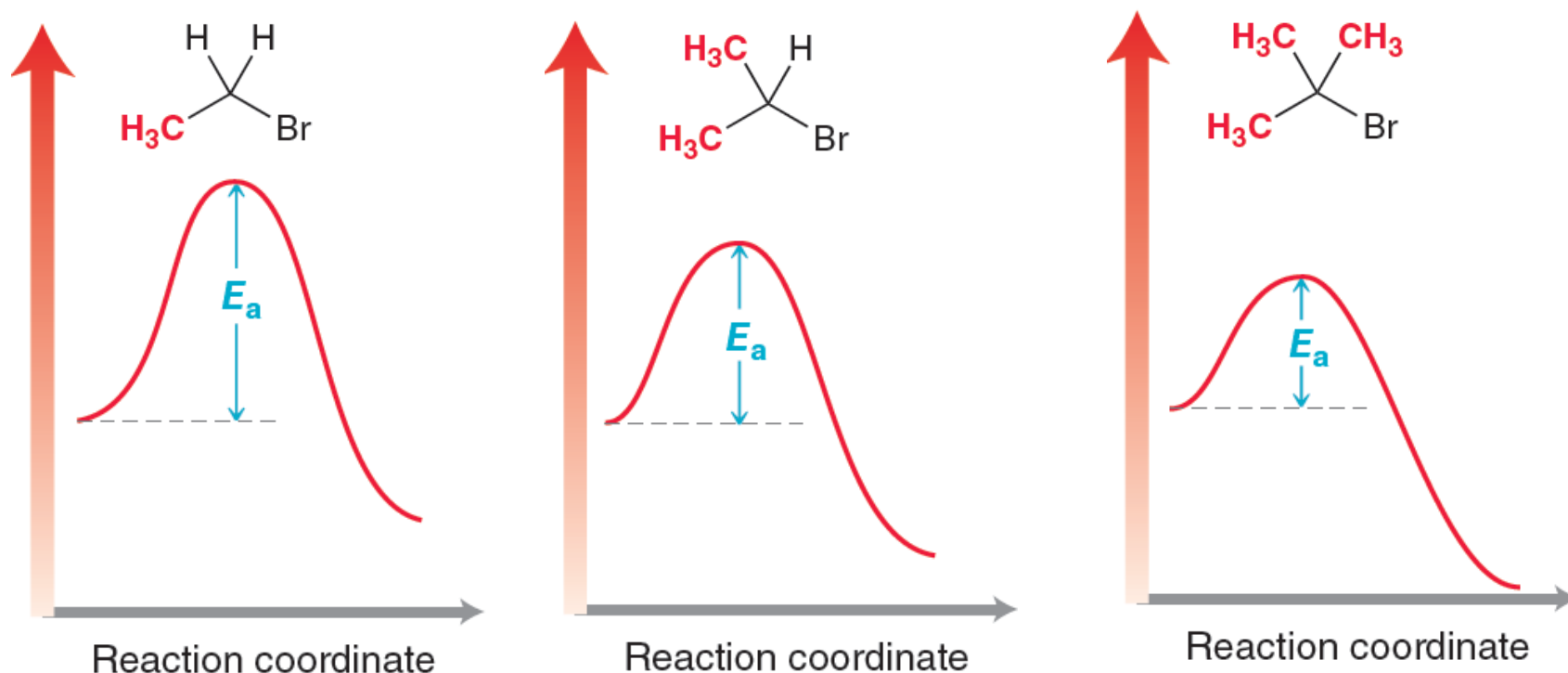
6.3 卤代烃的化学反应-消除反应

双分子消除历程 (E2)



6.3 卤代烃的化学反应-消除反应

双分子消除历程 (E2)



- 过渡态有部分双键性质，形成的双键取代越多，过渡态能量越低
- 一级烷基卤代物也可发生E2消除，只是速率比三级卤代物慢

6.3 卤代烃的化学反应-消除反应

Regioselectivity of E2 Reactions (区域选择性)

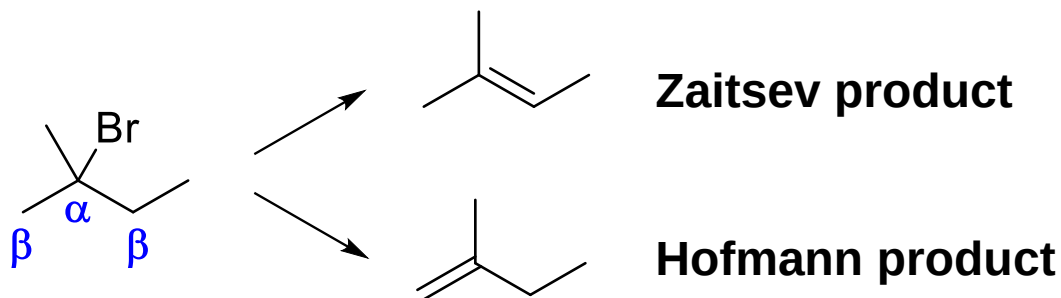
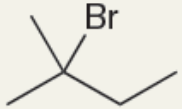
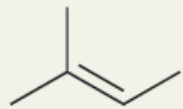
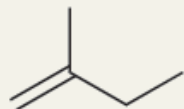
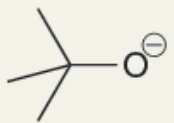


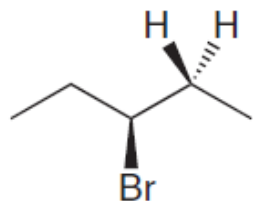
TABLE 8.1 PRODUCT DISTRIBUTION OF AN E2 REACTION AS A FUNCTION OF BASE

	Base $\rightarrow$		
		ZAITSEV	HOFMANN
EtO^-		71%	29%
		28%	72%
		8%	92%

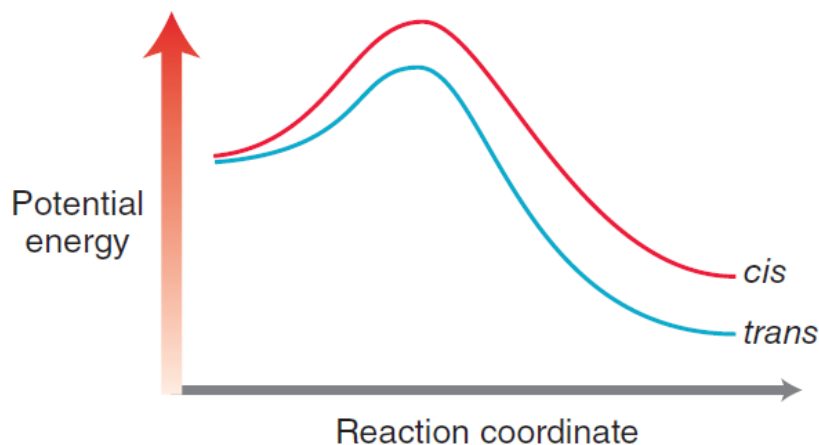
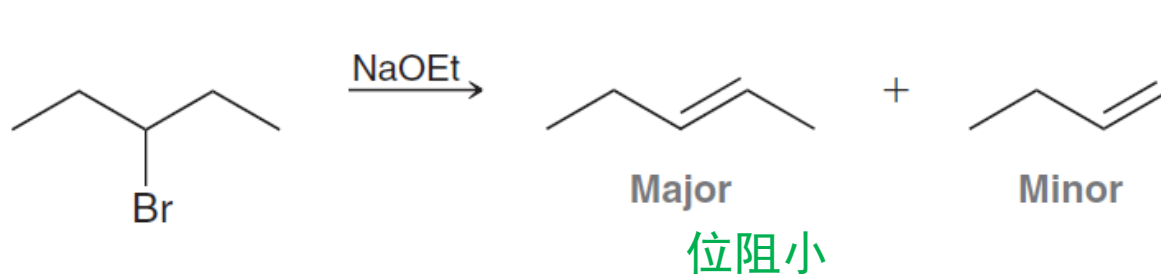
E2消除的区域选择性与碱的位阻有关

6.3 卤代烃的化学反应-消除反应

Stereoselectivity of E2 Reactions (立体选择性)



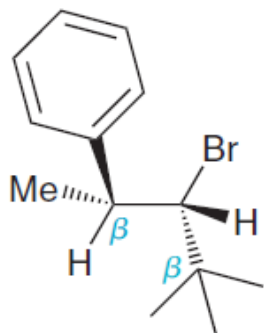
有2个 β -H, 可生成两种E2产物 **Stereoselectivity**



反应生成两种不等量的立体异构体，反应是具有立体选择性的

6.3 卤代烃的化学反应-消除反应

Stereospecificity of E2 Reactions (立体专一性)



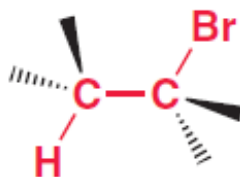
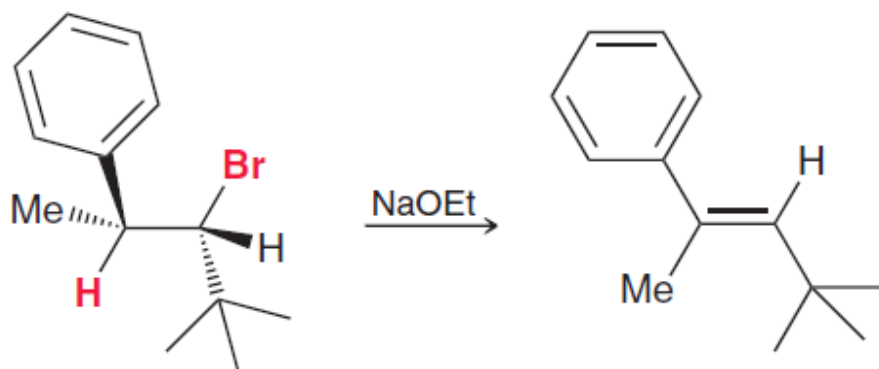
有1个 β -H, 只专一性的生成一种立体异构体 **Stereospecificity**

Why is the other possible stereoisomer not obtained?

6.3 卤代烃的化学反应-消除反应

Stereospecificity of E2 Reactions (立体专一性)

在发生E2消除的过程中形成 π 键，参与形成 π 键的四个原子需共平面。

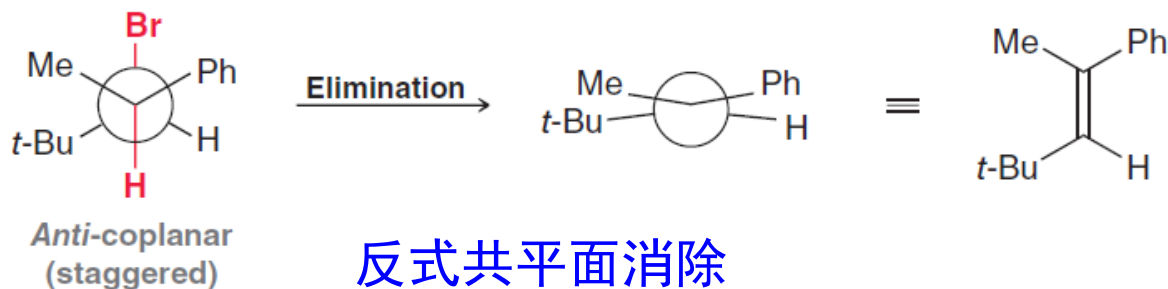
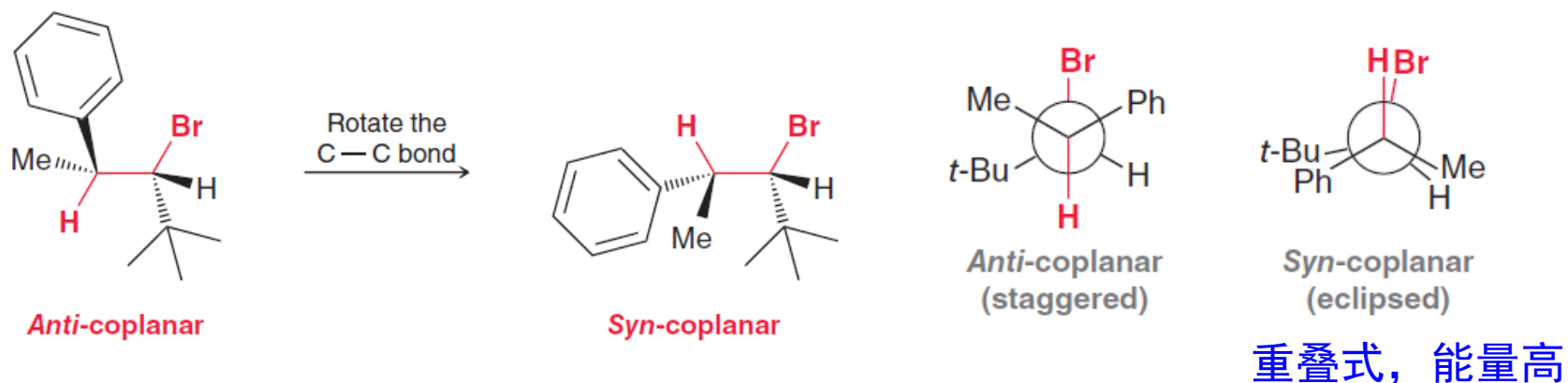


These four atoms (shown in red)
must all lie in the same plane

6.3 卤代烃的化学反应-消除反应

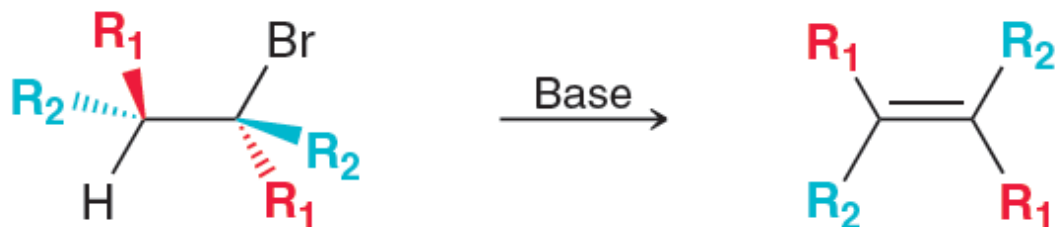
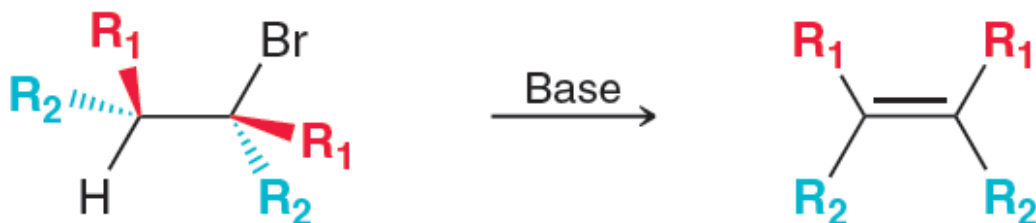
Stereospecificity of E2 Reactions (立体专一性)

两种可能的共平面方式：



6.3 卤代烃的化学反应-消除反应

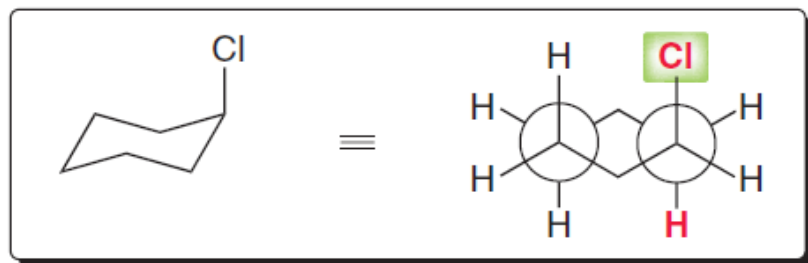
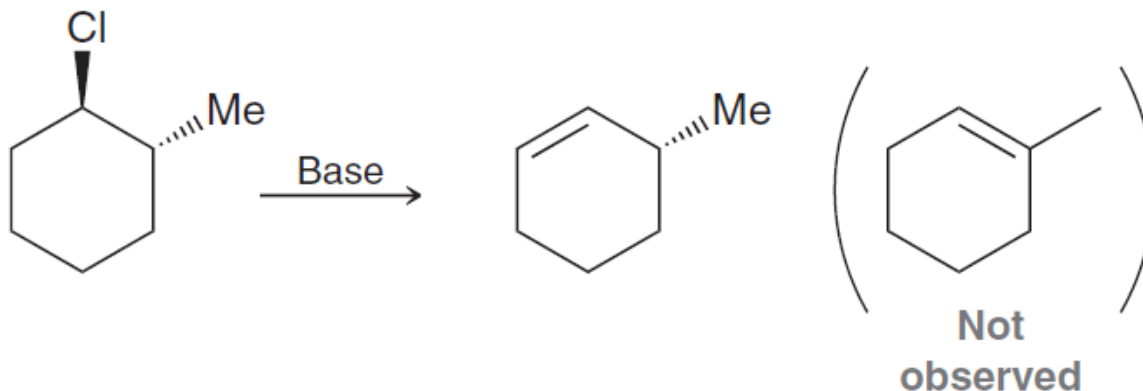
Stereospecificity of E2 Reactions (立体专一性)



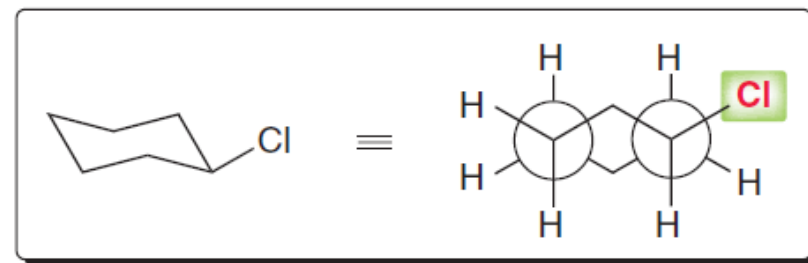
立体专一性：产物的立体构型取决于原料的立体构型

6.3 卤代烃的化学反应-消除反应

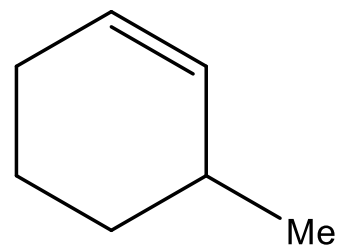
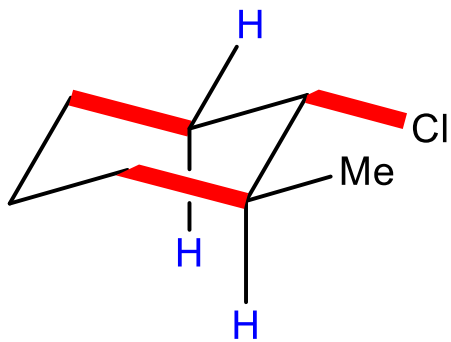
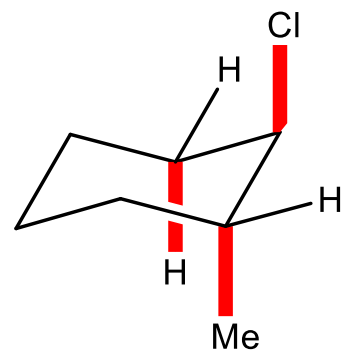
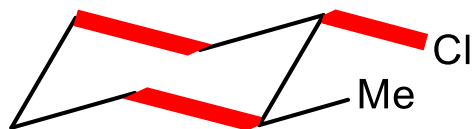
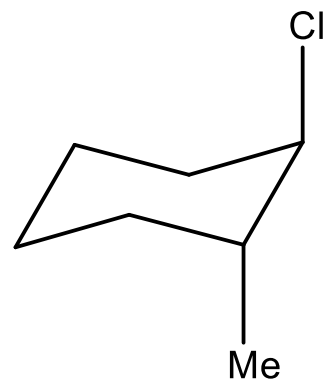
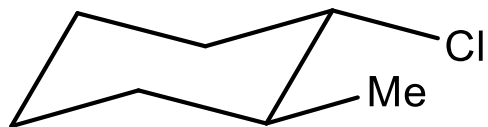
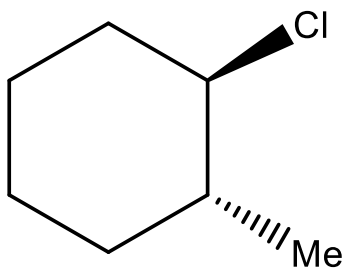
Stereospecificity of E2 Reactions (立体专一性)



When Cl is axial,
it can be *anti*-periplanar
with a neighboring hydrogen atom



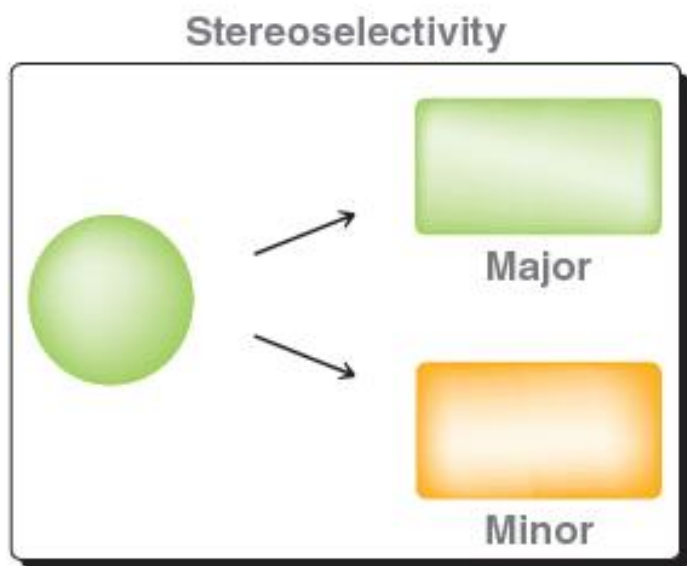
When Cl is equatorial,
it cannot be *anti*-periplanar with any
of its neighboring hydrogen atoms



6.3 卤代烃的化学反应-消除反应

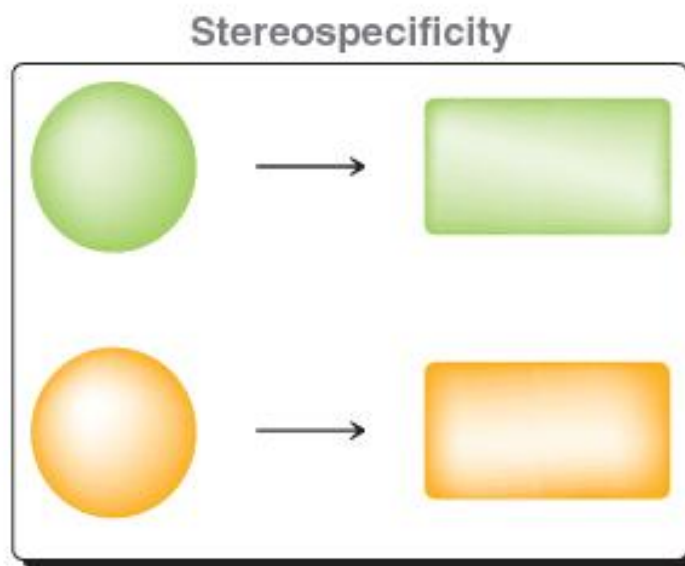
Stereoselectivity & Stereospecificity

立体选择性



从单一底物出发以一定的选择性得到两个可能的异构体

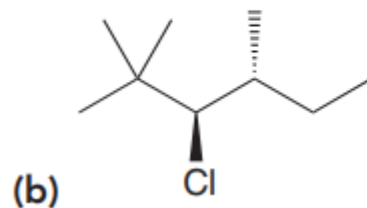
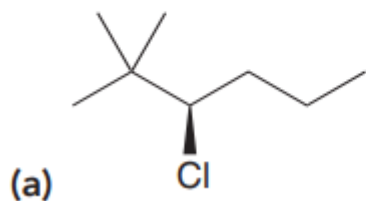
立体专一性



从两个互为异构体的底物出发得到两个互为异构体的产物

练习:

Identify the major and minor products for the E2 reaction that occurs when each of the following substrates is treated with a strong base:

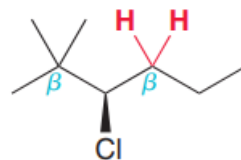


SOLUTION

(a) In this case, regiochemistry is irrelevant because there is only one β position that bears protons:

STEP 1

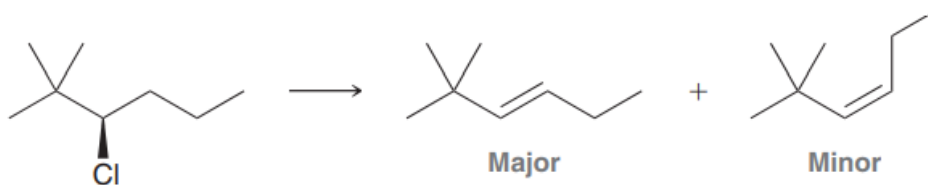
Identify all β positions with protons.



In order to determine if the reaction is stereoselective or stereospecific, count the number of protons at the β position. In this case, the β position has two protons. Therefore, we expect the reaction to be stereoselective. That is, we expect both *cis* and *trans* isomers, with a preference for *trans*:

STEP 2

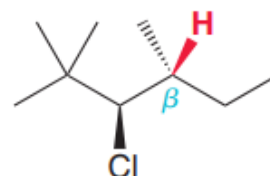
If the β position has two protons, expect *cis* and *trans* isomers.



- (b) In this case, the regiochemistry is also irrelevant, because there is once again only one β position that has protons:

STEP 1

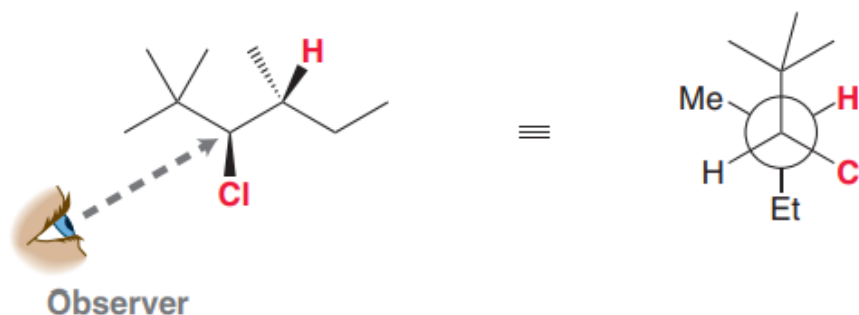
Identify all β positions with protons.



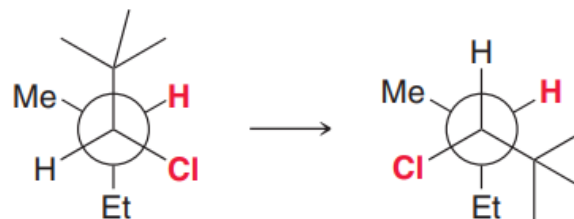
In order to determine if the reaction is stereoselective or stereospecific, count the number of protons at the β position. In this case, the β position has only one proton. Therefore, we expect the reaction to be stereospecific. That is, we expect only one particular stereoisomeric product, not both. To determine which product to expect, begin by drawing the Newman projection:

STEP 2

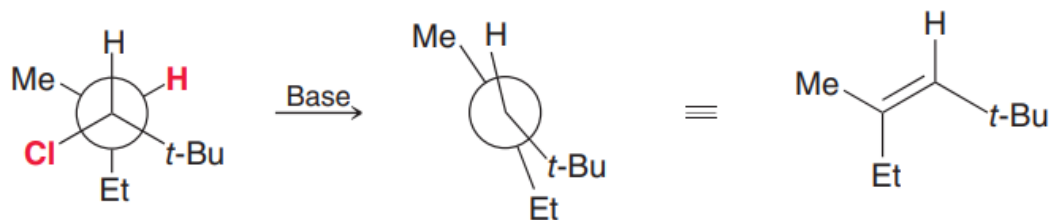
If the β position has only one proton, draw a Newman projection.



In this conformation, the proton and the leaving group (Cl) are not *anti*-periplanar. In order to achieve the appropriate conformation, rotate the C—C single bond so that the proton and the chlorine atom are *anti*-periplanar:



Now use this Newman projection to draw the product:



In this case, the product is the *Z* isomer. The *E* isomer is not obtained, because that would require elimination from a *syn*-periplanar conformation, which is eclipsed and too high in energy.

STEP 3

Rotate the C—C single bond to achieve an *anti*-periplanar conformation.

STEP 4

Use the Newman projection as a guide for drawing the product.

6.3 卤代烃的化学反应

亲核取代反应和消除反应的竞争

亲核取代反应和消除反应是伴随发生的，产物中同时存在消除产物和取代产物，二者比例受卤代烷的结构、进攻试剂的性质、温度、溶剂等多种因素的影响，通过控制反应条件可以使某种产物为主。

主要反应途径的判断

1. 判断试剂的功能：亲核性、碱性
2. 分析底物性质并判定反应机理
3. 分析产物的区域选择性、立体选择性

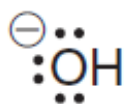
考验自己对化学知识的理解

认清本质，不要死记硬背！

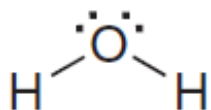
6.3 卤代烃的化学反应

试剂的碱性与亲核性

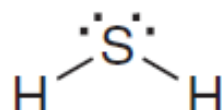
亲核性：电荷和可极化性



强亲核试剂



弱亲核试剂

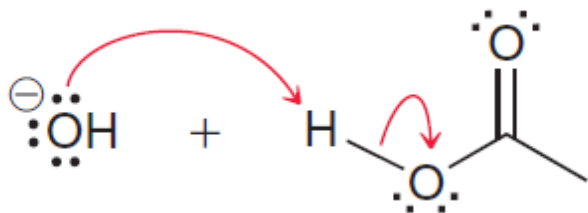


强亲核试剂

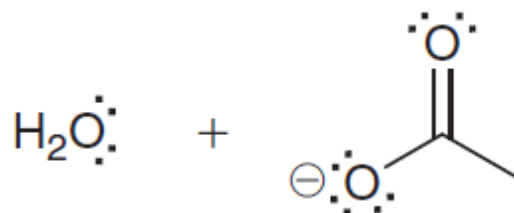


弱亲核试剂

碱性：化合物夺取质子的能力



强碱



弱碱

试剂的碱性与亲核性

**Nucleophile
(only)**

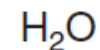
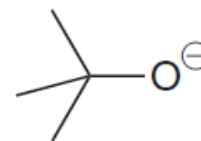
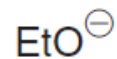
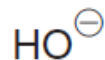
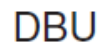
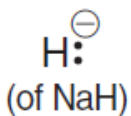
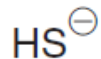
**Base
(only)**

**Strong Nuc / Strong
Base**

**Weak Nuc / Weak
Base**

Halides

Sulfur nucleophiles

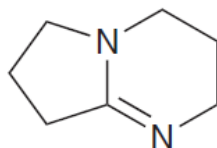


取代反应

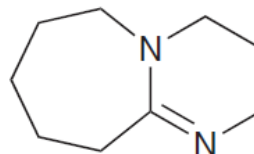
消除反应 (E2)

$\text{S}_{\text{N}}2$, E2

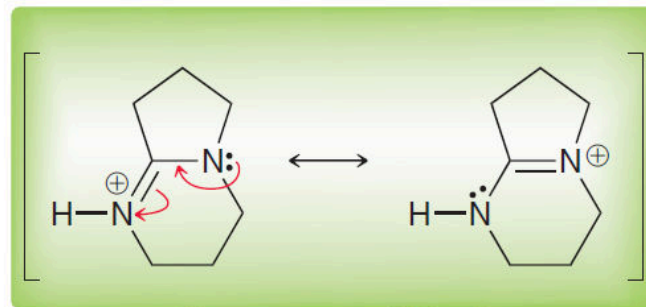
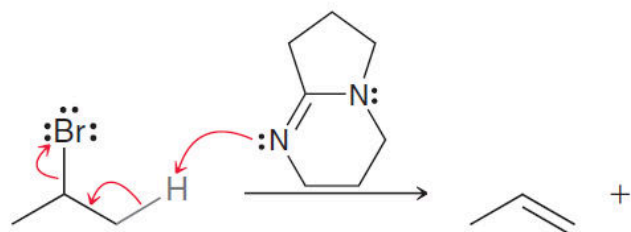
**$\text{S}_{\text{N}}1$, E1
 $\text{S}_{\text{N}}2$, E2**



1,5-Diazabicyclo[4.3.0]non-5-ene
(DBN)



1,8-Diazabicyclo[5.4.0]undec-7-ene
(DBU)



Resonance stabilized

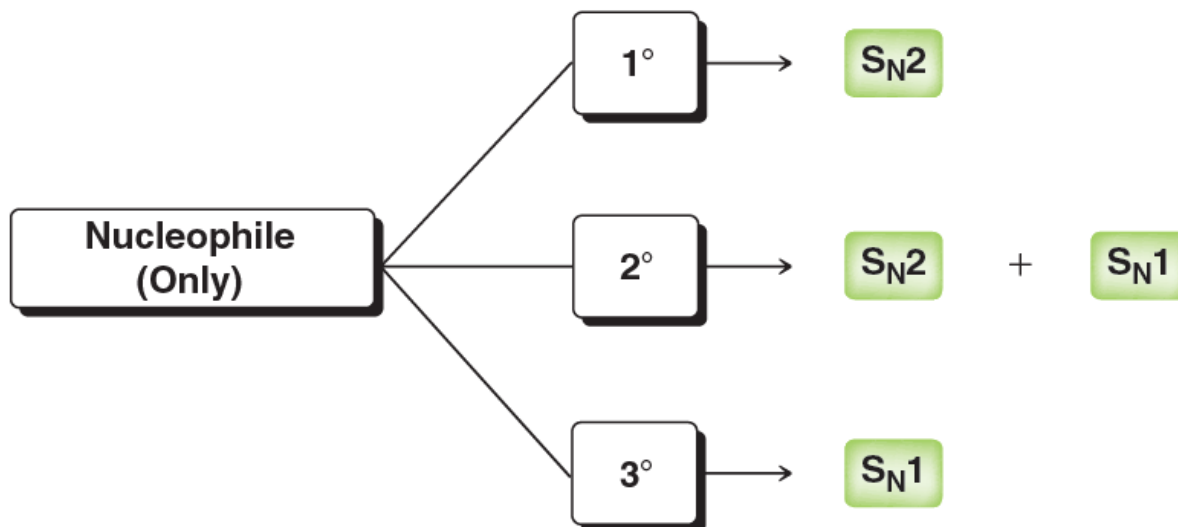


6.3 卤代烃的化学反应

第一类：仅做亲核试剂，取代反应

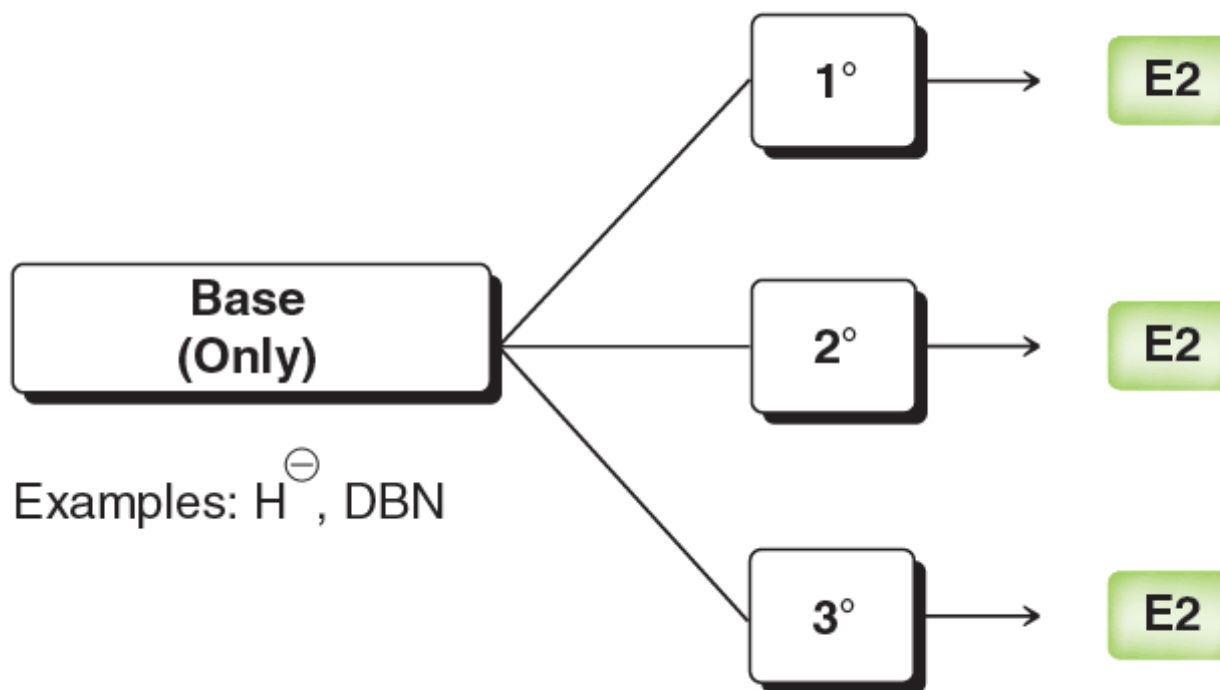
Halides	Sulfur nucleophiles	
$\text{Cl}^{\ominus}$	$\text{HS}^{\ominus}$	H_2S
$\text{Br}^{\ominus}$	$\text{RS}^{\ominus}$	RSH
$\text{I}^{\ominus}$		

1. 判断试剂的功能：亲核性、碱性
2. 分析底物性质并判定反应机理
3. 分析产物的区域选择性、立体选择性



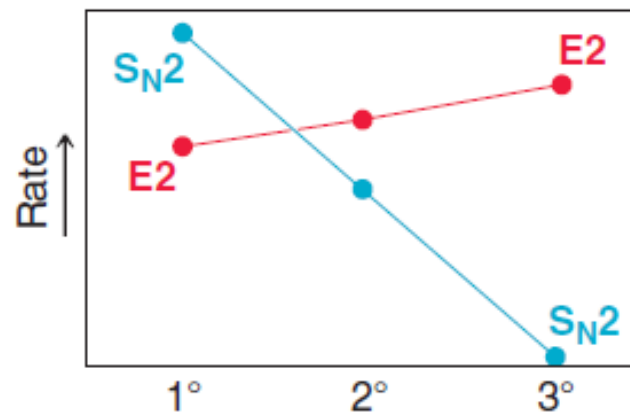
6.3 卤代烃的化学反应

第二类：仅做碱，消除反应

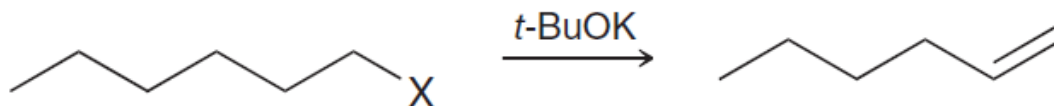


6.3 卤代烃的化学反应

第三类：强亲核试剂/强碱， $S_N2/E2$

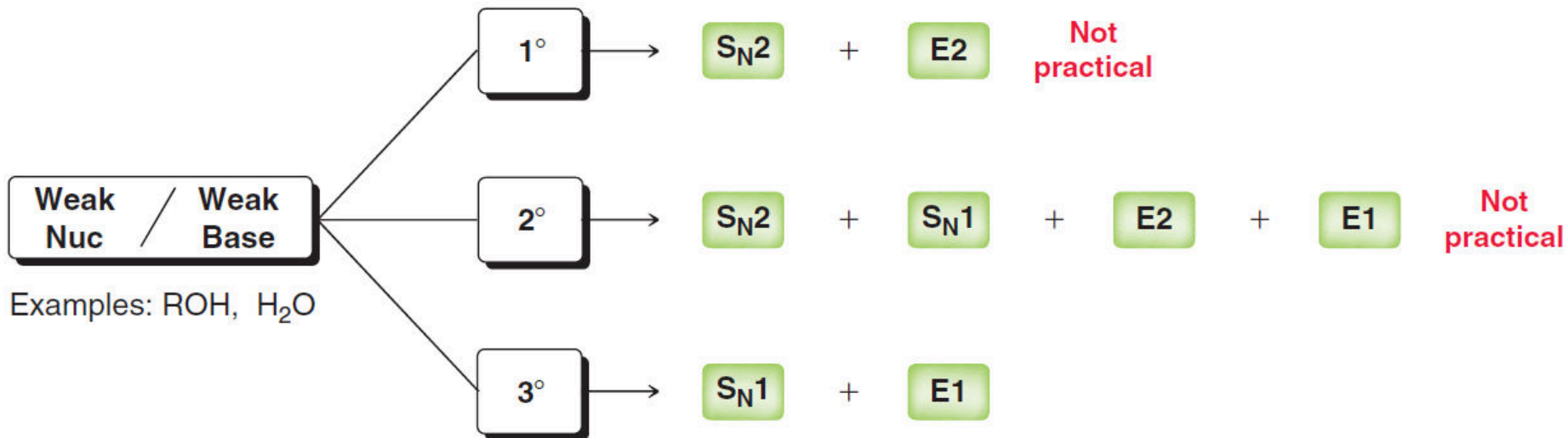


例外：大位阻碱有利于E2



6.3 卤代烃的化学反应

第四类：弱亲核试剂/弱碱， S_N2 / $E2$ / S_N1 / $E1$

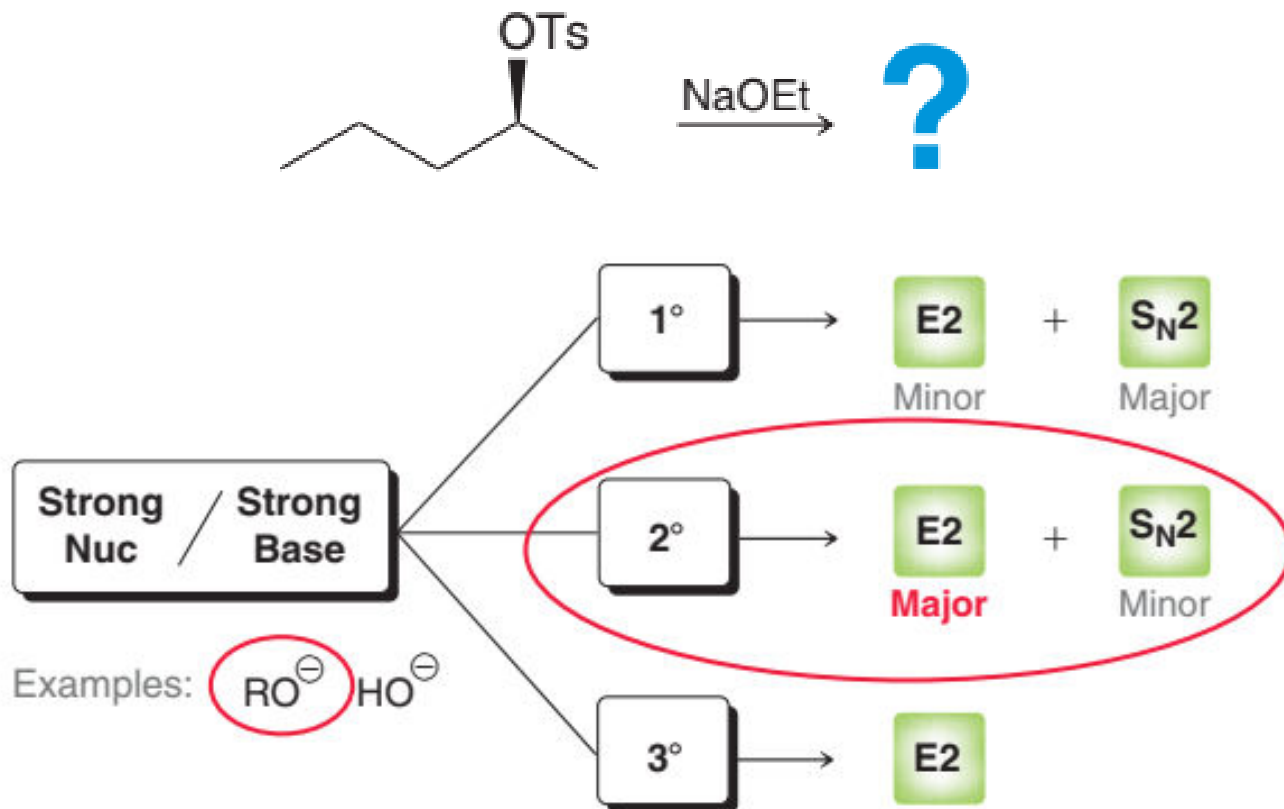


6.3 卤代烃的化学反应

1. 判断试剂的功能：亲核性、碱性
2. 分析底物性质并判定反应机理
3. 分析产物的区域选择性、立体选择性

	区域选择性	立体选择性
S_N2	进攻 α -C	构型翻转
S_N1	进攻 α -C	外消旋
E2	多取代烯烃为主；使用大位阻碱时会倾向少取代烯烃。	反式共平面消除 有两个 β -H (立体选择性) 有一个 β -H (立体专一性)
E1 (少)	多取代烯烃为主	反式烯烃为主

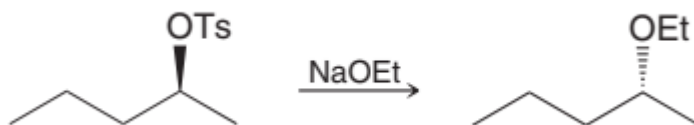
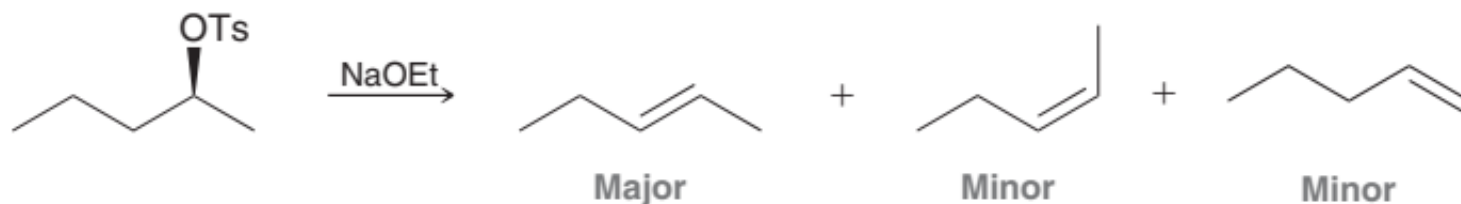
6.3 卤代烃的化学反应



1. 判断试剂的功能：亲核性、碱性
2. 分析底物性质并判定反应机理
3. 分析产物的区域选择性、立体选择性

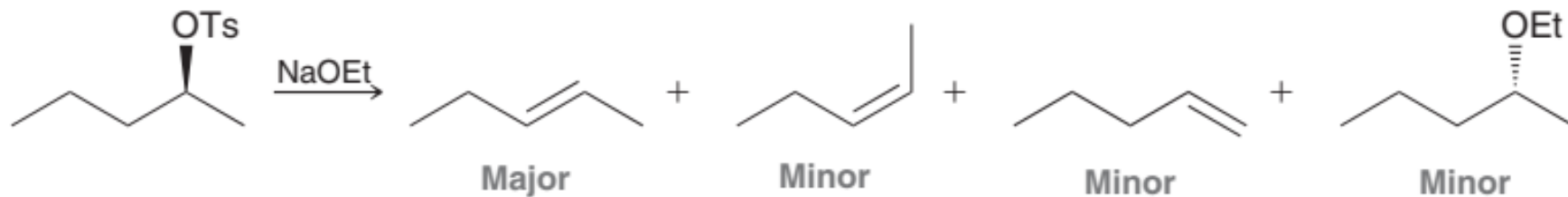
6.3 卤代烃的化学反应

1. 判断试剂的功能：亲核性、碱性
2. 分析底物性质并判定反应机理
3. 分析产物的区域选择性、立体选择性



6.3 卤代烃的化学反应

1. 判断试剂的功能：亲核性、碱性
2. 分析底物性质并判定反应机理
3. 分析产物的区域选择性、立体选择性

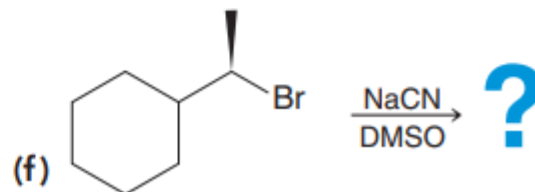
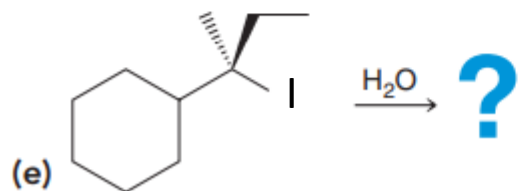
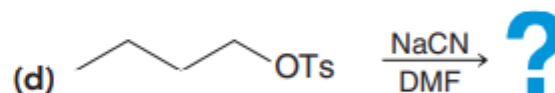
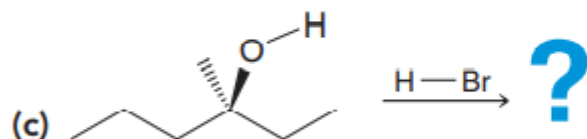
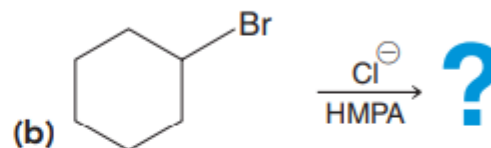
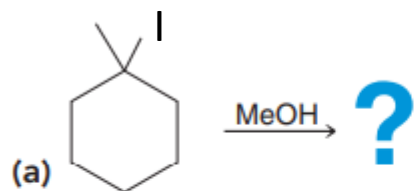


作业

手写!

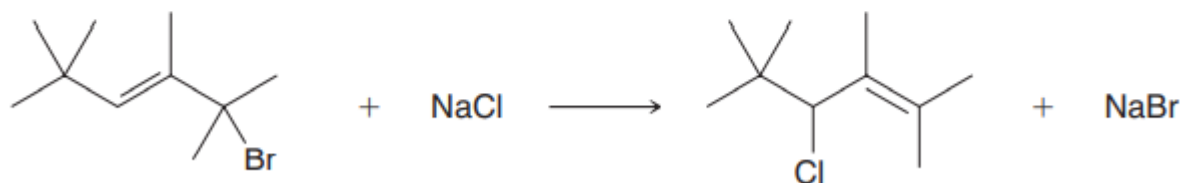
1

Determine whether each of the following reactions proceeds via an S_N1 or S_N2 mechanism and then draw the product(s) of the reaction:



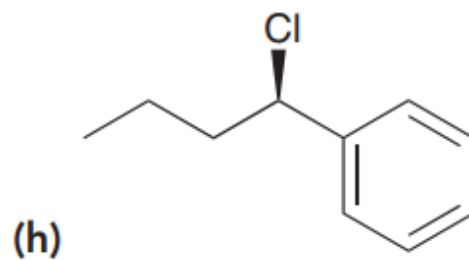
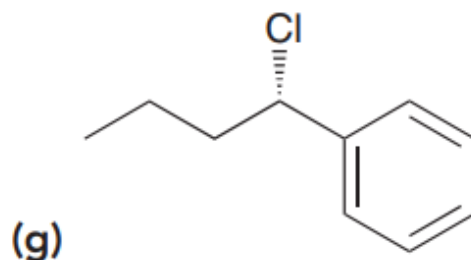
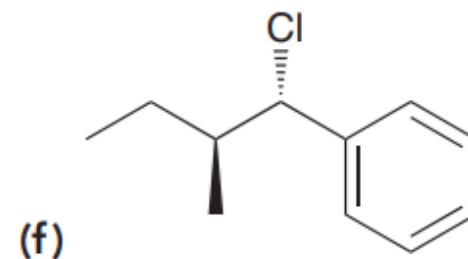
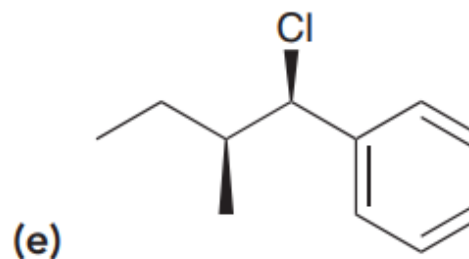
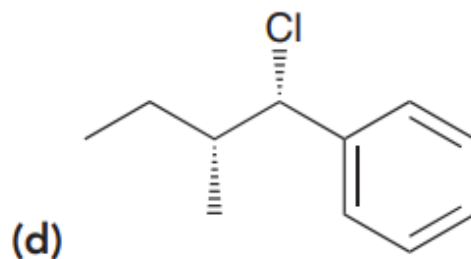
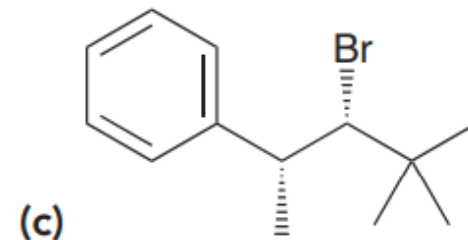
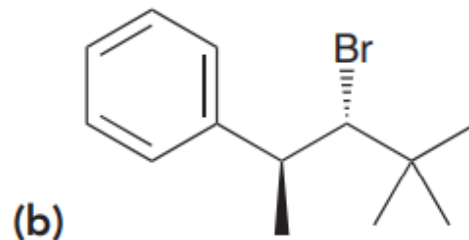
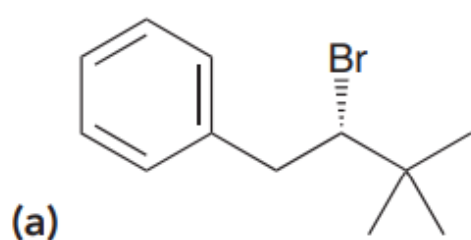
2

For the substitution reaction shown below, assume a stepwise process is taking place and draw the mechanism. (**Hint:** Review the rules for drawing resonance structures, Section 2.10.)



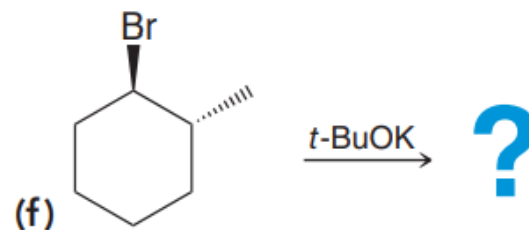
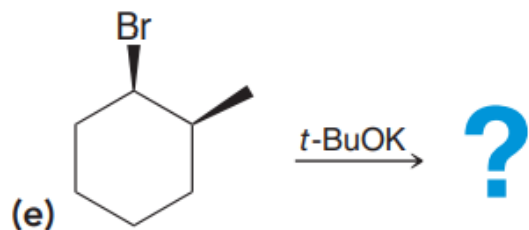
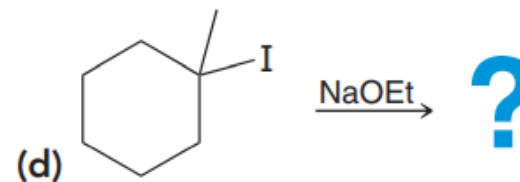
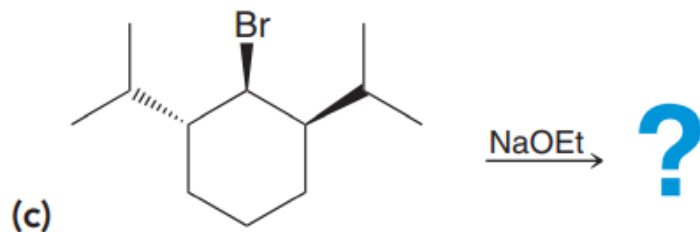
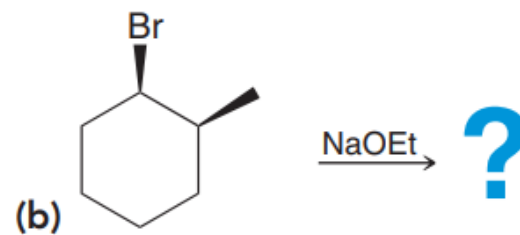
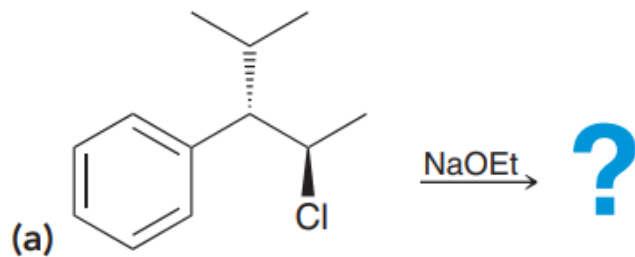
作业

3 Identify the major and minor products for the E2 reaction that occurs when each of the following substrates is treated with a strong base:



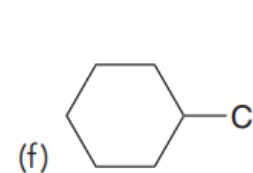
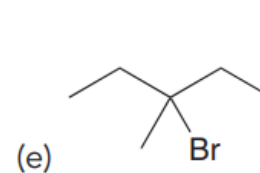
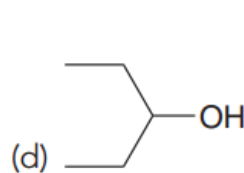
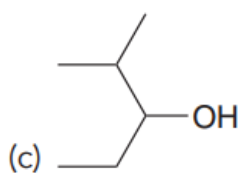
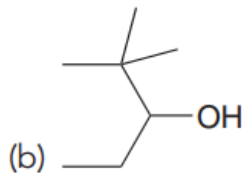
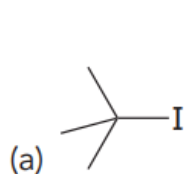
4

Predict the major and minor products for each of the following E2 reactions:



5

For each of the following substrates, determine whether an E1 process might involve a carbocation rearrangement or not:



6

Identify the major and minor product(s) that are expected for each of the following reactions:

